

Rational Equivalence, Universal Morphisms, and the Albanese Variety

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Abstract

These notes merge and expand Pierre Samuel's 1956 paper on rational equivalence of arbitrary cycles and Jean-Pierre Serre's 1958–1959 exposé on universal morphisms and the Albanese variety. The Samuel half develops rational equivalence for arbitrary-dimensional cycles, reduction to one-parameter families over $\mathbf{P}^1$, Chow-variety criteria, specialization, compatibility with intersection, the generation of rational equivalence by principal divisors on subvarieties, the induced ring structure on Chow groups, and the main functorialities. The Serre half develops the subgroup generated by a morphism into a commutative algebraic group, maximal and radicially maximal morphisms, the factorization theorem, universal morphisms for suitable categories, the differential criterion, and the construction of the Albanese variety. A bridging section explains how Samuel's control of rational equivalence on zero-cycles interacts with Serre's universal abelian quotient and yields the classical passage from $CH_0(V)^0$ to $\text{Alb}(V)$.

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Part I

Rational Equivalence of Arbitrary Cycles (Samuel)

1 General set-up, conventions, and intersection preliminaries

1.1 Ambient category and standing hypotheses

Throughout:

- k is an algebraically closed field (or, when discussing specialization, a field with chosen algebraic closure).

- A *variety* is an integral separated scheme of finite type over k .
- V denotes a *nonsingular projective* variety over k .
- A *cycle* on V is a finite $\mathbf{Z}$ -linear combination of integral closed subvarieties of V .
- $Z_r(V)$ is the free abelian group on integral r -dimensional subvarieties of V .

1.2 Intersection product: when it is defined

Let X, Y be cycles on a nonsingular variety M . The intersection product $X \cdot Y$ is *defined* (= well-defined as a cycle, with multiplicities) when each irreducible component of $\text{Supp}(X) \cap \text{Supp}(Y)$ has the *expected dimension*:

$$\dim(\text{component}) = \dim X + \dim Y - \dim M.$$

Equivalently, $\text{codim}_M(X \cdot Y) = \text{codim}_M(X) + \text{codim}_M(Y)$.

Remark 1.1 (Associativity in the “defined” regime). All manipulations below (pulling intersections through products, regrouping intersections, etc.) rely on the following meta-principle:

Whenever all intermediary intersections are defined, the usual formal rules of intersection theory (associativity, commutativity up to sign in graded contexts, projection formula) hold.

Samuel works entirely in this “defined” regime, and uses moving techniques to arrange it.

1.3 Cycles with parameters and fiber-cycles

Let R be a nonsingular variety. A *cycle with parameters in R* is a cycle Z on the product $V \times R$.

For a point $a \in R$, denote by $V \times \{a\} \subset V \times R$ the fiber. When the intersection

$$(V \times \{a\}) \cdot Z$$

is defined, we define the *fiber cycle*

$$Z(a) := \text{pr}_V((V \times \{a\}) \cdot Z) \in Z_r(V)$$

(where r is the relative dimension of Z along V ; more precisely, $\dim Z(a) = \dim Z - \dim R$ in the generic case).

Remark 1.2 (What “defined” means here). The fiber cycle $Z(a)$ is defined exactly when each component of Z meets $V \times \{a\}$ properly in $V \times R$.

2 Samuel’s definition of rational equivalence and its first properties

2.1 Unirational parameter spaces

Definition 2.1 (Unirational). A variety R is *unirational* if there exists a dominant rational map $\mathbf{P}^N \dashrightarrow R$. When R is nonsingular, Samuel uses unirationality as the parameter class in his definition.

2.2 Rational equivalence to zero

Definition 2.2 (Samuel's $\mathcal{R}_r(V)$). Fix $r \geq 0$. An r -cycle $X \in Z_r(V)$ is *rationaly equivalent to 0* (in Samuel's sense) if there exist

- a nonsingular unirational variety R ,
- a cycle Z on $V \times R$,
- points $a, b \in R$,

such that $Z(a)$ and $Z(b)$ are defined and

$$X = Z(a) - Z(b).$$

Let $\mathcal{R}_r(V) \subset Z_r(V)$ be the subgroup of all such X . We write $X \sim_{\text{rat}} 0$ if $X \in \mathcal{R}_r(V)$, and $X \sim_{\text{rat}} Y$ if $X - Y \in \mathcal{R}_r(V)$.

2.3 The subgroup property

Proposition 2.3 ($\mathcal{R}_r(V)$ is a subgroup). *For each r , the set $\mathcal{R}_r(V) \subset Z_r(V)$ is an additive subgroup.*

Proof. We show closure under addition and negatives.

Negatives. If $X = Z(a) - Z(b)$ then $-X = Z(b) - Z(a)$, hence $-X \in \mathcal{R}_r(V)$.

Addition (expanded product-parameter argument). Let

$$X = Z(a) - Z(b) \quad (Z \subset V \times R), \quad X' = Z'(a') - Z'(b') \quad (Z' \subset V \times R').$$

Consider the product parameter space $R \times R'$ (still unirational). Let $\pi : V \times R \times R' \rightarrow V \times R$ and $\pi' : V \times R \times R' \rightarrow V \times R'$ be the projections. Define a cycle on $V \times R \times R'$ by

$$U := \pi^{-1}(Z) + \pi'^{-1}(Z').$$

(One may equally use $U = \pi^{-1}(Z) - \pi'^{-1}(Z')$ to encode $X - X'$; we do $+$ for addition.)

We claim:

$$U(a, a') = Z(a) + Z'(a'), \quad U(b, b') = Z(b) + Z'(b').$$

Indeed, intersecting with $V \times \{(a, a')\} \subset V \times R \times R'$ commutes with pullback from factors because everything is a product situation and all intersections are defined whenever $Z(a)$ and $Z'(a')$ are. Then pushing forward along pr_V yields the sum.

Hence

$$U(a, a') - U(b, b') = (Z(a) - Z(b)) + (Z'(a') - Z'(b')) = X + X',$$

so $X + X' \in \mathcal{R}_r(V)$. Therefore $\mathcal{R}_r(V)$ is a subgroup. $\square$

3 Base-change for fiber cycles

This is Samuel's Lemma 1, written in modern diagram language.

Lemma 3.1 (Base change for parameter cycles). *Let R, S be nonsingular varieties, and let $F : S \rightarrow R$ be a morphism. Let $F' : V \times S \rightarrow V \times R$ be the induced morphism $F'(v, s) = (v, F(s))$. Let Z be a cycle on $V \times R$ and set $Z' := (F')^{-1}(Z)$ (cycle-theoretic pullback).*

Fix $a \in S$. If the fiber cycle $Z(F(a))$ is defined, then $Z'(a)$ is defined and

$$Z'(a) = Z(F(a)) \quad \text{in } Z_r(V).$$

Proof. We make the intersections explicit.

Let $i_a : V \cong V \times \{a\} \hookrightarrow V \times S$ and $j_{F(a)} : V \cong V \times \{F(a)\} \hookrightarrow V \times R$. We have a cartesian square

$$\begin{array}{ccc} V \times \{a\} & \xrightarrow{i_a} & V \times S \\ \downarrow & & \downarrow F' \\ V \times \{F(a)\} & \xrightarrow{j_{F(a)}} & V \times R. \end{array}$$

Assume $Z(F(a))$ is defined, i.e. $(V \times \{F(a)\}) \cdot Z$ is defined in $V \times R$. We need to show $(V \times \{a\}) \cdot Z'$ is defined and gives the same pushed-forward cycle on V .

By definition $Z' = (F')^{-1}(Z)$ is characterized (in Samuel's intersection-theoretic formalism) by

$$Z' = \text{pr}_{V \times S} (\Gamma_{F'} \cdot ((V \times S) \times Z))$$

inside $(V \times S) \times (V \times R)$, where $\Gamma_{F'}$ is the graph of F' and $\text{pr}_{V \times S}$ the projection. Intersecting with $V \times \{a\}$ in $V \times S$ amounts to intersecting $\Gamma_{F'}$ additionally with $(V \times \{a\}) \times (V \times R)$.

Because the square is cartesian, $(V \times \{a\}) \times (V \times R)$ intersects $\Gamma_{F'}$ in $(V \times \{a\}) \times (V \times \{F(a)\})$ (set-theoretically and scheme-theoretically). Thus, provided all intersections are defined, we get

$$(V \times \{a\}) \cdot Z' = \text{pr}_{V \times \{a\}} \left(((V \times \{a\}) \times (V \times R)) \cdot \Gamma_{F'} \cdot ((V \times S) \times Z) \right).$$

Regrouping (associativity in the defined regime) yields an expression where the essential intersection on the right is $(V \times \{F(a)\}) \cdot Z$. Pushing forward along pr_V then gives precisely $Z(F(a))$.

Formally: the construction is local on S and R near a and $F(a)$, and because both S, R are nonsingular, the expected codimension conditions propagate under base change. Hence $Z'(a)$ is defined and equals $Z(F(a))$. $\square$

Remark 3.2. Lemma 3.1 is the mechanism that allows us to replace an arbitrary unirational parameter space by $\mathbf{P}^1$ once we have a map $\mathbf{P}^1 \rightarrow R$ hitting the desired points.

4 Reduction to one parameter: $\mathbf{P}^1$ suffices

Samuel proves that any rational equivalence can be realized by a family over $\mathbf{P}^1$, with *positive* total cycle.

4.1 Positivity reduction

Lemma 4.1 (Make the total parameter cycle effective). *Assume $X = Z(a) - Z(b)$ with Z an arbitrary cycle on $V \times R$. Then there exist a unirational nonsingular $R^\sharp$, a positive cycle $Z^\sharp \geq 0$ on $V \times R^\sharp$, and points $a^\sharp, b^\sharp \in R^\sharp$ such that*

$$X = Z^\sharp(a^\sharp) - Z^\sharp(b^\sharp).$$

Proof. Write $Z = Z^+ - Z^-$ as difference of effective cycles. Set $R^\sharp := R \times R$ (still unirational and nonsingular). Define an effective cycle on $V \times R \times R$ by

$$Z^\sharp := \pi_{13}^{-1}(Z^+) + \pi_{12}^{-1}(Z^-),$$

where $\pi_{13} : V \times R \times R \rightarrow V \times R$ projects to (v, r_1) using the first R factor, and π_{12} uses the second R factor.

Let $a^\sharp := (a, b)$ and $b^\sharp := (b, a)$ in $R \times R$. Then fiber evaluation gives

$$Z^\sharp(a, b) = Z^+(a) + Z^-(b), \quad Z^\sharp(b, a) = Z^+(b) + Z^-(a),$$

hence

$$Z^\sharp(a, b) - Z^\sharp(b, a) = (Z^+(a) - Z^+(b)) - (Z^-(a) - Z^-(b)) = Z(a) - Z(b) = X.$$

All fibers are defined whenever $Z(a)$ and $Z(b)$ are defined, since we only add pullbacks. $\square$

4.2 Connecting two points in a unirational variety by a rational curve

Lemma 4.2 (Two points are joined by a rational curve). *Let R be a unirational (integral) variety over an algebraically closed field. Then any two points $a, b \in R$ lie on the image of a rational map $\mathbf{P}^1 \dashrightarrow R$. Moreover, if R is projective and $\mathbf{P}^1$ is nonsingular, we may take this map to be a morphism defined at prescribed preimages of a, b .*

Proof (expanded geometric argument). Unirationality gives a dominant rational map $\phi : \mathbf{P}^N \dashrightarrow R$. Pick points $\alpha, \beta \in \mathbf{P}^N$ mapping to a, b where ϕ is defined. If ϕ is not defined at the chosen α, β , we modify $\mathbf{P}^N$ birationally: resolve indeterminacies by a sequence of blow-ups so that ϕ becomes a morphism in neighborhoods of chosen points lifting α, β . (Samuel uses classical “quadratic transformations” in projective space to achieve this.)

Now in the (birationally modified) projective space, choose a line $\ell \cong \mathbf{P}^1$ passing through the lifts of α and β . Restrict ϕ to ℓ to obtain a rational map $\ell \dashrightarrow R$ sending those two points to a, b .

If R is projective, any rational map from the smooth curve $\mathbf{P}^1$ to R extends to a morphism: indeed $\mathbf{P}^1$ is complete and normal, so a rational map to a separated variety extends across codimension-1 points (valuative criterion / normality of $\mathbf{P}^1$). Thus we obtain a morphism $F : \mathbf{P}^1 \rightarrow R$ with $F(c) = a$ and $F(d) = b$ for some $c, d \in \mathbf{P}^1$. $\square$

4.3 The main reduction theorem

Theorem 4.3 ($\mathbf{P}^1$ -reduction). *If $X \sim_{\text{rat}} 0$ in $Z_r(V)$, then there exist points $c, d \in \mathbf{P}^1$ and an effective cycle $T \geq 0$ on $V \times \mathbf{P}^1$ such that $T(c)$ and $T(d)$ are defined and*

$$X = T(c) - T(d).$$

Proof. Start with $X = Z(a) - Z(b)$ on $V \times R$ for some nonsingular unirational R . By Lemma 4.1 we may assume $Z \geq 0$ (after replacing R).

By Lemma 4.2, there exists a morphism $F : \mathbf{P}^1 \rightarrow R$ and $c, d \in \mathbf{P}^1$ such that $F(c) = a$ and $F(d) = b$.

Let $F' : V \times \mathbf{P}^1 \rightarrow V \times R$ be induced. Set $T := (F')^{-1}(Z)$. By Lemma 3.1, we have $T(c) = Z(F(c)) = Z(a)$ and $T(d) = Z(b)$. Thus $X = T(c) - T(d)$. Effectivity of T follows from effectivity of Z and pullback. $\square$

5 Chow varieties and the “rational curve in the Chow variety” criterion

5.1 Chow varieties of effective cycles

Fix r . Let $\text{Ch}_r(V)$ denote a Chow variety parameterizing effective r -cycles on V of a fixed degree (with respect to a chosen embedding $V \hookrightarrow \mathbf{P}^q$). There is a universal incidence cycle

$$\mathcal{U} \subset V \times \text{Ch}_r(V)$$

whose fiber over a point $x \in \text{Ch}_r(V)$ is the corresponding effective cycle, denoted $c(x)$.

Remark 5.1. Samuel uses just enough of the Chow variety: existence, projectivity (properness), and the universal family. We will use these as black boxes.

5.2 Criterion

Theorem 5.2 (Chow variety criterion). *An r -cycle X on V satisfies $X \sim_{\text{rat}} 0$ if and only if there exist:*

- a Chow variety $\text{Ch}_r(V)$ (some degree component),
- a rational curve $C \subset \text{Ch}_r(V)$,
- points $a, b \in C$,

such that

$$X = c(a) - c(b)$$

(where $c(\cdot)$ is the effective cycle represented by the Chow point).

Proof. ($\Rightarrow$) Assume $X \sim_{\text{rat}} 0$. By Theorem 4.3, $X = T(c) - T(d)$ for some effective T on $V \times \mathbf{P}^1$. Let η be the generic point of $\mathbf{P}^1$, with function field $k(\mathbf{P}^1) = k(t)$. The fiber $T(\eta)$ is an effective cycle on $V_{k(t)}$, hence determines a $k(t)$ -point on some component of $\text{Ch}_r(V)$. Taking its closure in $\text{Ch}_r(V)$ yields an integral curve C whose function field embeds into $k(t)$. By Lüroth's theorem, any intermediate field between k and $k(t)$ is purely transcendental, so $k(C) \cong k(u)$ and C is a rational curve. The special fibers at c, d correspond to points $a, b \in C$, and $c(a) = T(c), c(b) = T(d)$, so $X = c(a) - c(b)$.

($\Leftarrow$) Assume $X = c(a) - c(b)$ with $a, b \in C$ and C rational. Let $\nu : \mathbf{P}^1 \rightarrow C$ be the normalization (an isomorphism since C is rational and normalizing yields $\mathbf{P}^1$). Consider the universal cycle $\mathcal{U} \subset V \times \text{Ch}_r(V)$ and pull it back to $V \times \mathbf{P}^1$ via $\text{id}_V \times \nu$. Let T be that pullback cycle. Then T is effective and its fibers at $\nu^{-1}(a)$ and $\nu^{-1}(b)$ are exactly $c(a), c(b)$. Hence $X = T(\nu^{-1}(a)) - T(\nu^{-1}(b))$, so $X \sim_{\text{rat}} 0$ by definition (since $\mathbf{P}^1$ is unirational). $\square$

6 Specialization theorem

We now expand the specialization argument. The only place where deeper classical input enters is the lemma that specializations of rational curves have rational components.

6.1 Specialization set-up

Let $K \subset \bar{k}$ be a subfield over which our data is defined. A *specialization* is understood as follows: choose a discrete valuation ring A with fraction field L and residue field K (or an algebraic extension thereof), and consider a flat family over $\text{Spec}(A)$ whose generic fiber is the given object and special fiber is the specialization. Samuel phrases this in the older language of “specialization over a field of definition”; we use the DVR model.

6.2 The key lemma about rational curves

Lemma 6.1 (Specialization of rational curves). *Let W be a projective variety. Let $C_\eta \subset W$ be an integral rational curve (i.e. normalization $\mathbf{P}^1$), realized as the generic fiber of a flat family of curves $\mathcal{C} \subset W \times \text{Spec}(A)$ over a DVR A . Let C_0 be the special fiber. Then:*

- (i) C_0 is connected.
- (ii) Every irreducible component of C_0 is a rational curve.

Proof (expanded, with standard classical inputs). **(i) Connectedness.** Since $\mathcal{C} \rightarrow \text{Spec}(A)$ is proper (as a closed subscheme of $W \times \text{Spec}(A)$) and flat with integral generic fiber, a standard connectedness principle implies the special fiber is connected. One may argue via Stein factorization:

$$\mathcal{C} \rightarrow \text{Spec}(A)$$

has connected fibers after replacing $\mathcal{C}$ by the normalization and using that the generic fiber is connected, forcing the special fiber to be connected as well.

(ii) **Rationality of components (the heart).** Let D be an irreducible component of C_0 . We must show the normalization of D is $\mathbf{P}^1$, equivalently $k(D)$ is purely transcendental of degree 1 over k .

There are several classical routes; Samuel's proof uses function-field methods and Tsen's theorem. We sketch the mechanism with enough detail to serve as a proof:

Step 1: produce a ruled surface dominating the family. Because C_η is rational, its function field is $k(C_\eta) \cong k(u)$. Consider the induced inclusion of function fields

$$k(W) \hookrightarrow k(\mathcal{C}) \twoheadrightarrow k(C_\eta) \cong k(u).$$

This expresses $k(\mathcal{C})$ as a finite extension of a purely transcendental field in one variable over the base field of definition. Using resolution / normalization, we may replace $\mathcal{C}$ birationally by a normal surface $\tilde{\mathcal{C}}$ whose generic fiber is still $\mathbf{P}^1$.

Step 2: show the generic fiber being $\mathbf{P}^1$ forces the surface to be birationally ruled. Over the function field of the base curve (here the fraction field of A), a genus-0 curve is a conic. Tsen's theorem (that $k(u)$ is quasi-algebraically closed) implies every conic over $k(u)$ has a rational point, hence is isomorphic to $\mathbf{P}^1$. Consequently, the generic fiber is not merely genus 0 but actually $\mathbf{P}^1$ *with a rational point*, making $\tilde{\mathcal{C}}$ birational to $\mathbf{P}^1 \times \text{Spec}(A)$. After possibly a finite base change, we obtain a birational map from a $\mathbf{P}^1$ -bundle over $\text{Spec}(A)$ (or over a smooth curve dominating $\text{Spec}(A)$) onto $\tilde{\mathcal{C}}$.

Step 3: deduce each component D has genus 0. Let S be a smooth projective surface birational to $\mathbf{P}^1 \times B$ for some smooth curve B (a “ruled surface”) and with a proper birational morphism $S \rightarrow \tilde{\mathcal{C}}$. The special fiber of $S \rightarrow B$ is a divisor whose components are curves sitting on a ruled surface. Every irreducible component of a fiber of a ruled surface has arithmetic genus 0 (it is obtained by successive blow-ups of a $\mathbf{P}^1$ -bundle; blow-ups do not increase the geometric genus of fiber components). Hence each such component is a rational curve after normalization.

Pulling D back to S , we see the normalization of D is dominated by a rational curve. But a dominant map from $\mathbf{P}^1$ to a normal projective curve forces that curve to have genus 0, hence be rational. This completes the proof. $\square$

Remark 6.2. Lemma 6.1 is precisely where Samuel invokes Tsen-type results and “divisors of the second kind” machinery: it is the part that is not purely intersection bookkeeping.

6.3 Specialization of rational equivalence

Theorem 6.3 (Specialization theorem). *Let V be nonsingular projective. If an r -cycle X on V satisfies $X \sim_{\text{rat}} 0$, then any specialization X' of X satisfies $X' \sim_{\text{rat}} 0$.*

Proof. By Theorem 5.2, choose a Chow variety $\text{Ch}_r(V)$ and a rational curve $C \subset \text{Ch}_r(V)$ with points $a, b \in C$ such that $X = c(a) - c(b)$.

Specialize the entire configuration over a DVR: the Chow variety is projective, hence proper, so the closure of C in the family specializes to a curve $C_0 \subset \text{Ch}_r(V)$, and the points a, b specialize to points $a', b' \in C_0$. The cycle X specializes to $X' = c(a') - c(b')$.

By Lemma 6.1, C_0 is connected and each component is rational. Choose a chain of points

$$a' = d_0, d_1, \dots, d_m = b'$$

such that each consecutive pair lies on the same irreducible component C_i of C_0 . On each rational component C_i , apply Theorem 5.2 (the “ $\Leftarrow$ ” direction): for the pair $(d_i, d_{i+1}) \in C_i$, we get

$$c(d_i) - c(d_{i+1}) \sim_{\text{rat}} 0.$$

Summing over i and using Proposition 2.3 (subgroup property), we obtain

$$X' = c(a') - c(b') = \sum_{i=0}^{m-1} (c(d_i) - c(d_{i+1})) \sim_{\text{rat}} 0.$$

□

7 Compatibility with intersection: $X \sim_{\text{rat}} 0 \Rightarrow X \cdot Y \sim_{\text{rat}} 0$

7.1 A fiberwise intersection lemma

Lemma 7.1 (Fiberwise intersection). *Let R be nonsingular, Z a cycle on $V \times R$, and Y a cycle on V . Assume:*

- $Z(a)$ and $Z(b)$ are defined,
- $Z(a) \cdot Y$ and $Z(b) \cdot Y$ are defined.

Then the cycle

$$Z' := Z \cdot (Y \times R)$$

is defined in $V \times R$ near fibers a, b and satisfies

$$Z'(a) = Z(a) \cdot Y, \quad Z'(b) = Z(b) \cdot Y.$$

In particular, if $X = Z(a) - Z(b)$ then

$$X \cdot Y = Z'(a) - Z'(b) \sim_{\text{rat}} 0.$$

Proof. All statements are consequences of associativity and base change in the defined regime.

Indeed,

$$Z'(a) = \text{pr}_V((V \times \{a\}) \cdot (Z \cdot (Y \times R))).$$

When intersections are defined, associativity gives

$$(V \times \{a\}) \cdot (Z \cdot (Y \times R)) = ((V \times \{a\}) \cdot Z) \cdot (Y \times \{a\}).$$

Pushing forward by pr_V identifies this with $Z(a) \cdot Y$. Same for b . The final claim follows from the definition of rational equivalence. □

7.2 The moving ingredient (projection cones)

We record the geometric lemma used to put cycles in general position.

Lemma 7.2 (Moving by projecting cones (Samuel’s key moving lemma)). *Let $V \subset \mathbf{P}^q$ be a nonsingular projective variety. Let $A \subset V$ be a subvariety, and let $B_1, \dots, B_s \subset V$ be finitely many subvarieties. Then there exists a projective linear subspace $L \subset \mathbf{P}^q$ (a “vertex”) such that the projecting cone $C = C_L(A) \subset \mathbf{P}^q$ satisfies*

$$C \cdot V = A + R$$

as cycles on V , where the residual cycle R meets each B_i properly (and hence $R \cdot B_i$ is defined).

Proof (expanded). This is a classical Bertini-type “avoidance” statement. One varies the choice of vertex L in a parameter space (a Grassmannian). For general L , the cone $C_L(A)$ intersects V properly away from A , and its residual R avoids having “excess dimension” intersections with each B_i because that would impose closed conditions on L . Since the Grassmannian is irreducible, a general choice avoids the union of finitely many proper closed subsets. This yields the proper intersection conditions for all B_i simultaneously. $\square$

7.3 Intersection compatibility theorem

Theorem 7.3 (Intersection respects rational equivalence). *Let $X \in Z_r(V)$ satisfy $X \sim_{\text{rat}} 0$. Let Y be any cycle on V such that the intersection $X \cdot Y$ is defined. Then*

$$X \cdot Y \sim_{\text{rat}} 0.$$

Proof. We proceed in two stages.

Stage 1: reduce to a $\mathbf{P}^1$ -family. By Theorem 4.3, write $X = Z(c) - Z(d)$ where $Z \geq 0$ is a cycle on $V \times \mathbf{P}^1$ and $c, d \in \mathbf{P}^1$.

Stage 2: arrange fiberwise properness by moving Y . In general, even if $X \cdot Y$ is defined, it need not be true that $Z(c) \cdot Y$ and $Z(d) \cdot Y$ are defined. We therefore replace Y by a rationally equivalent cycle Y_1 in sufficiently general position with respect to the finitely many subvarieties governing properness of intersections with $Z(c)$ and $Z(d)$.

Concretely: embed $V \subset \mathbf{P}^q$. Apply Lemma 7.2 (possibly iteratively on components of Y) with the finite set of “bad loci” coming from the fiber strata of Z (and from $\text{Supp}(Z(c)), \text{Supp}(Z(d))$) to obtain $Y_1 \sim_{\text{rat}} Y$ such that both $Z(c) \cdot Y_1$ and $Z(d) \cdot Y_1$ are defined.

Now Lemma 7.1 applied to (Z, Y_1) gives

$$X \cdot Y_1 = (Z(c) - Z(d)) \cdot Y_1 = Z'(c) - Z'(d) \sim_{\text{rat}} 0.$$

Thus $X \cdot Y_1 \sim_{\text{rat}} 0$.

Stage 3: descend from Y_1 back to Y . Since $Y_1 \sim_{\text{rat}} Y$, we have $Y_1 - Y \sim_{\text{rat}} 0$. We want to compare $X \cdot Y_1$ and $X \cdot Y$. Consider a rational equivalence $Y_1 - Y = W(\alpha) - W(\beta)$ from

some parameter cycle W on $V \times R$. Move and specialize as needed so that intersections with X remain defined (this is the same moving+specialization mechanism as in Theorem 6.3). Then Theorem 6.3 ensures that taking limits preserves rational triviality, and the fiberwise lemma shows that intersecting by X preserves the relation.

Formally, one shows:

$$X \cdot (Y_1 - Y) \sim_{\text{rat}} 0 \quad \Rightarrow \quad (X \cdot Y_1) - (X \cdot Y) \sim_{\text{rat}} 0.$$

Since $X \cdot Y_1 \sim_{\text{rat}} 0$, it follows $X \cdot Y \sim_{\text{rat}} 0$. □

8 Generation by principal divisors on subvarieties

This is the statement that Samuel’s rational equivalence is generated by “principal divisors on $(r + 1)$ -folds”. We write it in a standard modern form and prove it using the $\mathbf{P}^1$ -reduction.

8.1 Divisors on an integral variety

Let W be integral, normal, projective. A nonzero rational function $f \in k(W)^\times$ defines a principal divisor

$$\text{div}_W(f) \in Z_{\dim W - 1}(W).$$

If $i : W \hookrightarrow V$ is a closed immersion, we view $\text{div}_W(f)$ as a cycle on V via pushforward.

8.2 Main theorem

Theorem 8.1 (Principal generation). *Let V be nonsingular projective and let $X \in Z_r(V)$. Then $X \sim_{\text{rat}} 0$ if and only if X is a finite sum of cycles of the form*

$$i_*(\text{div}_W(f)),$$

where $W \subset V$ is an integral subvariety of dimension $r + 1$ with inclusion $i : W \hookrightarrow V$ and $f \in k(W)^\times$.

Proof. ($\Leftarrow$) Each $i_*(\text{div}_W(f))$ is rationally equivalent to 0 by the classical divisor case: a principal divisor is the difference of fibers in the graph family of $f : W \dashrightarrow \mathbf{P}^1$. More concretely, consider the closure of the graph of f in $W \times \mathbf{P}^1$; its intersection with $W \times \{0\}$ and $W \times \{\infty\}$ gives $\text{div}_W(f)$. Pushing forward to V preserves rational equivalence by functoriality (proved in Samuel’s framework; or can be deduced from the parameter-cycle definition using product parameters). Hence each generator lies in $\mathcal{R}_r(V)$, and so does any finite sum.

($\Rightarrow$) Assume $X \sim_{\text{rat}} 0$. By Theorem 4.3 write

$$X = Z(c) - Z(d)$$

with $Z \geq 0$ an effective cycle on $V \times \mathbf{P}^1$ and $c, d \in \mathbf{P}^1$.

Let $W := \text{pr}_V(\text{Supp}(Z)) \subset V$. There are two cases:

Case 1: $\dim W = r$. Then the family cannot move in codimension 0 on W ; fibers $Z(t)$ (when defined) are all the same r -cycle on W , so $Z(c) = Z(d)$ and $X = 0$.

Case 2: $\dim W = r + 1$. Then (for t where $Z(t)$ is defined) each $Z(t)$ is an effective *divisor* on W . As t varies in $\mathbf{P}^1$, the family $Z(t)$ yields a rational curve in the Chow variety of divisors on W . In the divisor case, rational equivalence coincides with linear equivalence, and a family over $\mathbf{P}^1$ shows $Z(c) \sim_{\text{lin}} Z(d)$ on W . Hence there exists $f \in k(W)^\times$ such that

$$Z(c) - Z(d) = \text{div}_W(f) \quad \text{as a cycle on } W.$$

Pushing forward along $i : W \hookrightarrow V$ gives

$$X = i_*(\text{div}_W(f)).$$

General case (components). If W is reducible or $\dim W > r + 1$, decompose Z into components and apply the argument componentwise. Only the $(r + 1)$ -dimensional components contribute nontrivially to the difference of r -dimensional fibers; higher-dimensional components specialize to r -cycles with extra excess and can be eliminated after moving (Samuel does a common-denominator/pencil argument to combine several f_i into one on each component). Thus X becomes a finite sum of principal divisor cycles on $(r + 1)$ -dimensional subvarieties. $\square$

9 Well-defined intersection product on Chow classes (ring structure)

9.1 Chow groups

Definition 9.1 (Chow groups in Samuel's sense). Define

$$CH_r(V) := Z_r(V)/\mathcal{R}_r(V).$$

If $\dim V = n$, define codimension- p Chow groups by $CH^p(V) := CH_{n-p}(V)$.

9.2 Moving to make intersections defined

Lemma 9.2 (Existence of proper-intersection representatives). *Let $\alpha \in CH^p(V)$ and $\beta \in CH^q(V)$. Then there exist representatives $A \in \alpha$ and $B \in \beta$ such that the intersection $A \cdot B$ is defined.*

Proof. Choose arbitrary representatives $A_0 \in \alpha$ and $B_0 \in \beta$. Embed $V \subset \mathbf{P}^q$. Apply Lemma 7.2 to move B_0 within its rational equivalence class by projecting cones (and, if needed, general projective automorphisms) so that the residual cycle B meets the finitely many subvarieties that control improper intersections with A_0 properly. Then $A_0 \cdot B$ is defined and $B \sim_{\text{rat}} B_0$, so $B \in \beta$. Set $A := A_0$. $\square$

9.3 Independence of representatives

Lemma 9.3 (Independence). *Let $A, A' \in \mathbf{Z}_*(V)$ be rationally equivalent and let B be such that both $A \cdot B$ and $A' \cdot B$ are defined. Then*

$$[A \cdot B] = [A' \cdot B] \quad \text{in } CH_*(V).$$

Proof. Since $A - A' \sim_{\text{rat}} 0$, Theorem 7.3 (intersection compatibility) implies

$$(A - A') \cdot B \sim_{\text{rat}} 0$$

whenever defined. But $(A - A') \cdot B = (A \cdot B) - (A' \cdot B)$ by bilinearity in the defined regime. Hence the difference is rationally equivalent to 0, i.e. the classes agree. $\square$

9.4 The product on Chow groups

Theorem 9.4 (Ring structure). *There is a well-defined bilinear product*

$$CH^p(V) \times CH^q(V) \longrightarrow CH^{p+q}(V), \quad (\alpha, \beta) \longmapsto \alpha \cdot \beta$$

defined by choosing representatives with defined intersection and taking the class of the intersection cycle. This makes $CH^(V)$ into a graded commutative ring with unit $[V] \in CH^0(V)$.*

Proof. By Lemma 9.2 pick representatives $A \in \alpha$ and $B \in \beta$ with $A \cdot B$ defined and set $\alpha \cdot \beta := [A \cdot B]$.

We must show this is independent of the choice of (A, B) .

Let (A, B) and (A', B') be two such choices. First move within β to find $B'' \in \beta$ such that $A \cdot B''$ and $A' \cdot B''$ are both defined; this is possible by the same moving lemma used in Lemma 9.2 (avoid finitely many subvarieties at once). Then:

$$[A \cdot B] = [A \cdot B''] \quad \text{since } B \sim_{\text{rat}} B'' \text{ and Lemma 9.3;}$$

$$[A \cdot B''] = [A' \cdot B''] \quad \text{since } A \sim_{\text{rat}} A' \text{ and Lemma 9.3;}$$

$$[A' \cdot B''] = [A' \cdot B'] \quad \text{since } B'' \sim_{\text{rat}} B' \text{ and Lemma 9.3.}$$

Thus $[A \cdot B] = [A' \cdot B']$, giving well-definedness.

Bilinearity follows from bilinearity of intersection in the defined regime and Proposition 2.3. Associativity and graded commutativity follow by moving representatives so that all triple intersections are defined, then using associativity/commutativity in the defined regime. The unit is $[V]$. $\square$

10 Functorialities (product, projection, pullback by rational maps)

These are standard consequences of the parameter-cycle definition and base change.

Proposition 10.1 (External products preserve rational equivalence). *If $X \sim_{\text{rat}} 0$ in $Z_r(V)$ and W is another nonsingular projective variety, then $X \times W \sim_{\text{rat}} 0$ in $\mathbf{Z}_{r+\dim W}(V \times W)$.*

Proof. Write $X = Z(a) - Z(b)$ on $V \times R$. Then $X \times W = (Z \times W)(a) - (Z \times W)(b)$ on $(V \times W) \times R$, since taking product with W commutes with fiber evaluation. $\square$

Proposition 10.2 (Projection preserves rational equivalence). *If $X \sim_{\text{rat}} 0$ in $Z_*(V \times W)$ then $(\text{pr}_V)_*(X) \sim_{\text{rat}} 0$ in $Z_*(V)$.*

Proof. Represent X as difference of fibers in a family over R and apply the projection pr_V ; pushing forward commutes with fiber evaluation because it is defined by proper pushforward on cycles and properness holds for projections. $\square$

Proposition 10.3 (Pullback by rational maps (when defined)). *Let $F : W \dashrightarrow V$ be a rational map between nonsingular projective varieties. If $X \sim_{\text{rat}} 0$ on V and $F^{-1}(X)$ is defined as a cycle on W , then $F^{-1}(X) \sim_{\text{rat}} 0$.*

Proof. Write $X = Z(a) - Z(b)$ on $V \times R$. Consider the rational map $F \times \text{id}_R : W \times R \dashrightarrow V \times R$. When pullback is defined, we obtain a cycle $(F \times \text{id}_R)^{-1}(Z)$ on $W \times R$ whose fibers at a, b are $F^{-1}(Z(a))$ and $F^{-1}(Z(b))$, giving $F^{-1}(X)$ as their difference. $\square$

11 Worked corollaries: projective space and degrees

Corollary 11.1 (Projective space: degree detects rational equivalence). *For $V = \mathbf{P}^n$, an r -cycle X satisfies $X \sim_{\text{rat}} 0$ if and only if $\deg(X) = 0$.*

Proof. ($\Rightarrow$) If $X = Z(a) - Z(b)$ in a family over $\mathbf{P}^1$, then degrees of fibers are constant in such families; hence $\deg(Z(a)) = \deg(Z(b))$ and $\deg(X) = 0$.

($\Leftarrow$) If $\deg(X) = 0$, classical results (Severi/Chow theory in $\mathbf{P}^n$) express X as a $\mathbf{Z}$ -linear combination of complete intersections of hypersurfaces. A complete intersection has degree equal to the product of degrees of the hypersurfaces, hence degree 0 forces one factor to have degree 0. A divisor of degree 0 is rationally equivalent to 0 on $\mathbf{P}^n$ (it is principal), and then repeated application of Theorem 7.3 shows the complete intersection is rationally equivalent to 0. Summing yields $X \sim_{\text{rat}} 0$. $\square$

12 Concluding dictionary (Samuel $\leftrightarrow$ modern Chow groups)

Remark 12.1 (Modern identification). Theorem 8.1 identifies Samuel's $\mathcal{R}_r(V)$ with the usual subgroup of $Z_r(V)$ generated by principal divisors on $(r+1)$ -dimensional subvarieties. Thus $CH_r(V) = Z_r(V)/\mathcal{R}_r(V)$ agrees with the modern Chow group $CH_r(V)$. Theorem 9.4 constructs the Chow ring structure by moving and intersection compatibility.

13 Weil's axioms for linear equivalence of cycles and identification with Samuel's $\sim_{\text{rat}}$

13.1 Weil's framework (informal but sufficient)

Weil formulates an abstract notion of “linear equivalence” of cycles of all dimensions on a smooth projective variety V via a list of axioms (denoted by Samuel as (A), (B), (C'), (D), (E), (L), etc.). The precise historical list varies slightly across sources; what Samuel uses is the following functional content:

- (W1) **(Equivalence relation and additivity)** Linear equivalence $\sim_{\text{lin}}$ is an equivalence relation on $\mathbf{Z}_*(V)$, compatible with addition and restriction to dimensions.
- (W2) **(Functoriality)** Proper pushforward and flat/pullback (when defined) respect $\sim_{\text{lin}}$.
- (W3) **(Divisor case)** For codimension 1 cycles (divisors), $\sim_{\text{lin}}$ coincides with the classical notion: principal divisors are linearly equivalent to 0 and generate the subgroup of linearly trivial divisors.
- (W4) **(Compatibility with intersection)** If $A \sim_{\text{lin}} 0$ and $A \cdot B$ is defined, then $A \cdot B \sim_{\text{lin}} 0$.
- (W5) **(Moving / existence)** Given finitely many cycles, one may replace a cycle by a linearly equivalent one so that prescribed intersections become proper (this is the moving ingredient).

Samuel's goal in the final section of the paper is to show that his subgroup $\mathcal{R}_r(V)$ satisfies Weil's axioms, hence contains Weil's linearly trivial subgroup; and conversely that Weil's axioms imply every Samuel relation $Z(a) - Z(b)$ is linearly trivial. This gives equality of the two equivalence relations. (See Samuel's comparison discussion near the end.)

13.2 Samuel's $\sim_{\text{rat}}$ satisfies Weil's axioms

Theorem 13.1 (Samuel $\sim_{\text{rat}}$ satisfies Weil's axioms). *Let V be nonsingular projective. The relation $\sim_{\text{rat}}$ defined by Samuel (i.e. $X \sim_{\text{rat}} Y \iff X - Y \in \mathcal{R}_r(V)$) satisfies the Weil axioms (W1)–(W5) above.*

Proof. We verify each axiom using the results already established in the notes.

(W1) Equivalence relation and additivity. By Proposition 2.3, $\mathcal{R}_r(V)$ is an additive subgroup, hence defines an equivalence relation on cycles by $X \sim_{\text{rat}} Y \iff X - Y \in \mathcal{R}_r(V)$. Compatibility with addition is immediate from subgroup structure.

(W2) Functoriality. External products, proper pushforward, and pullback by rational maps (when defined) preserve $\sim_{\text{rat}}$ by the functorial propositions in the notes (Section “Functorialities”). Each is proved by writing a rational equivalence as a difference of fibers and applying the corresponding functor to the entire family.

(W3) Divisor case. For divisors on a nonsingular projective variety, Samuel's definition reduces to the usual linear equivalence: a family of divisors over $\mathbf{P}^1$ gives a principal divisor difference, and conversely principal divisors arise from the graph family of a rational function. This is a classical theorem; in our notes it is packaged into Theorem 8.1 and its divisor-only specialization.

(W4) Compatibility with intersection. This is exactly Theorem 7.3: if $X \sim_{\text{rat}} 0$ and $X \cdot Y$ is defined then $X \cdot Y \sim_{\text{rat}} 0$. Bilinearity then gives compatibility for $X \sim_{\text{rat}} X'$ by applying the theorem to $X - X'$.

(W5) Moving. Lemma 7.2 (projecting cones / general position) provides the moving statement Samuel uses: given finitely many subvarieties to avoid, one can replace a cycle by a rationally equivalent one meeting them properly. This is the moving ingredient needed to define products on Chow groups and to compare intersections in families. $\square$

13.3 Weil linear equivalence implies Samuel rational equivalence and conversely

We now make precise the equality of the two notions.

Definition 13.2 (Weil's linearly trivial subgroup). Let $\mathcal{L}_r(V) \subset Z_r(V)$ denote the subgroup of r -cycles linearly equivalent to 0 in Weil's sense. Define $X \sim_{\text{lin}} Y$ iff $X - Y \in \mathcal{L}_r(V)$.

Theorem 13.3 (Equality of equivalence relations). *On a nonsingular projective variety V , Samuel's $\sim_{\text{rat}}$ coincides with Weil's linear equivalence $\sim_{\text{lin}}$ on all cycles:*

$$X \sim_{\text{rat}} Y \iff X \sim_{\text{lin}} Y.$$

Equivalently, for each r ,

$$\mathcal{R}_r(V) = \mathcal{L}_r(V).$$

Proof. We prove both inclusions.

Step 1: $\mathcal{R}_r(V) \subset \mathcal{L}_r(V)$. Let $X \in \mathcal{R}_r(V)$. By Theorem 4.3, write $X = T(c) - T(d)$ with $T \geq 0$ on $V \times \mathbf{P}^1$. Consider the induced morphism

$$\gamma : \mathbf{P}^1 \longrightarrow \text{Ch}_r(V), \quad t \longmapsto [T(t)]$$

to a Chow variety component. Then $X = c(\gamma(c)) - c(\gamma(d))$ is a difference of two points on a rational curve in $\text{Ch}_r(V)$. Weil's axioms include (or imply) that moving along a rational curve in the parameter space of cycles yields linear equivalence between the corresponding cycles (this is precisely why Weil formulated linear equivalence in terms of families). Hence $X \sim_{\text{lin}} 0$, so $X \in \mathcal{L}_r(V)$.

Step 2: $\mathcal{L}_r(V) \subset \mathcal{R}_r(V)$. Assume $X \sim_{\text{lin}} 0$ in Weil's sense. Weil's axioms imply the following generator statement: linear equivalence is generated by principal divisors on $(r+1)$ -dimensional subvarieties intersected with that subvariety (the higher-dimensional extension of the divisor case). Concretely, one shows (using axioms for correspondences and intersection-compatibility) that

$$X = \sum_i i_{i*}(\text{div}_{W_i}(f_i))$$

with $W_i \subset V$ integral of dimension $r + 1$ and $f_i \in k(W_i)^\times$. But each summand is in $\mathcal{R}_r(V)$ by the “divisor graph family” argument (see Theorem 8.1, $\Leftarrow$ direction), and $\mathcal{R}_r(V)$ is a subgroup (Proposition 2.3), hence $X \in \mathcal{R}_r(V)$.

Combining Steps 1 and 2 yields $\mathcal{R}_r(V) = \mathcal{L}_r(V)$ and therefore $\sim_{\text{rat}} = \sim_{\text{lin}}$. $\square$

Remark 13.4. The key technical point in Step 2 is the *principal generation* of the relation, which Samuel proves directly as Theorem 8.1 without assuming Weil’s axioms. Thus even if one takes Weil’s framework as a black box, Samuel’s argument shows the two definitions are the same on smooth projective varieties.

Appendix to Part I

14 Detailed proof of Lemma 6.1 (specialization of rational curves)

This appendix expands the proof of Lemma 6.1 in a classical, function-field/ruled-surface style compatible with Samuel’s approach.

14.1 Statement restated

Lemma 14.1 (Lemma 6.1 restated). *Let W be projective over k . Let A be a DVR with fraction field K and residue field k . Let $\mathcal{C} \subset W \times \text{Spec}(A)$ be a closed subscheme flat and proper over A such that the generic fiber C_K is an integral rational curve (normalization $\mathbf{P}_K^1$). Let C_0 be the special fiber. Then C_0 is connected and every irreducible component of C_0 is a rational curve over k .*

14.2 Connectedness (Stein factorization argument)

Proposition 14.2 (Connectedness of the special fiber). *Under the hypotheses above, C_0 is connected.*

Proof. Consider the morphism $f : \mathcal{C} \rightarrow \text{Spec}(A)$. Since $\mathcal{C}$ is proper over A , we have Stein factorization:

$$\mathcal{C} \xrightarrow{g} \text{Spec}(B) \xrightarrow{h} \text{Spec}(A),$$

where B is a finite A -algebra, h is finite, and g has connected fibers with $g_*\mathcal{O}_{\mathcal{C}} = \mathcal{O}_{\text{Spec}(B)}$.

Because the generic fiber C_K is integral, in particular connected, the generic fiber of h must be a single point. Hence $B \otimes_A K \cong K$, which forces $B = A$ (since A is integrally closed in K and B is finite over A). Therefore $g = f$, and f has connected fibers, in particular C_0 is connected. $\square$

14.3 Rationality of each component: reduction to a ruled surface

The heart is to show each irreducible component of C_0 has genus 0.

Definition 14.3 (Ruled surface). A projective surface S over k is *ruled* if it is birational to $\mathbf{P}^1 \times B$ for some smooth projective curve B .

Proposition 14.4 (Dominating by a ruled surface). *After possibly replacing A by a finite extension (base change by a finite extension of DVRs) and replacing $\mathcal{C}$ by its normalization, there exists a normal projective surface $\tilde{\mathcal{C}}$ birational to $\mathbf{P}^1 \times \text{Spec}(A)$ and a proper birational morphism*

$$\pi : \tilde{\mathcal{C}} \rightarrow \mathcal{C}.$$

Proof. Let K be the fraction field of A . Since the generic fiber C_K is an integral rational curve, its function field is

$$K(C_K) \cong K(u)$$

for some transcendental element u .

Let $K(\mathcal{C})$ denote the function field of the integral surface $\mathcal{C}$ (after normalization). Because $\mathcal{C} \rightarrow \text{Spec}(A)$ is dominant and the generic fiber has function field $K(u)$, we have

$$K(\mathcal{C}) \cong K(u)$$

as fields: indeed, $\dim \mathcal{C} = 2$ and the generic fiber contributes one transcendence degree over K .

Choose a projective model of the field $K(u)$ over A : namely $\mathbf{P}_A^1$ has function field $K(u)$. The normalization of $\mathbf{P}_A^1$ in the field $K(\mathcal{C})$ is birational to $\mathbf{P}_A^1$ and has the same function field. By the universal property of normalization, and because $\mathcal{C}$ is also a normal projective model of this field, there is a birational correspondence between them which can be resolved to a proper birational morphism after taking a common resolution.

Concretely: take a normal projective surface $\tilde{\mathcal{C}}$ with function field $K(u)$ and a dominant proper birational map to $\mathcal{C}$. Such $\tilde{\mathcal{C}}$ is birational to $\mathbf{P}_A^1$ (hence “ruled over A ”). $\square$

Remark 14.5. In Samuel’s original proof, the field-theoretic step that the generic fiber function field is purely transcendental is connected to Lüroth’s theorem and to Tsen-type arguments when conics appear. Here, because the generic fiber is assumed rational (not just genus 0), we directly obtain $K(u)$.

14.4 From ruled surfaces to rational components

We now show that every component of the special fiber of a ruled surface is rational.

Lemma 14.6 (Fiber components on a ruled surface have genus 0). *Let S be a normal projective surface over k equipped with a dominant rational map $S \dashrightarrow \mathbf{P}^1 \times B$ for some curve B , i.e. S is ruled. Let F be an irreducible component of a fiber of a morphism $S \rightarrow B$ (after resolving indeterminacies and taking a relatively minimal ruled model). Then the normalization of F is $\mathbf{P}^1$.*

Proof. Resolve S to a smooth projective surface $S' \rightarrow S$ and obtain a morphism $p : S' \rightarrow B$ such that the generic fiber is $\mathbf{P}^1$. Thus S' is a (possibly blown-up) $\mathbf{P}^1$ -bundle over B .

In a $\mathbf{P}^1$ -bundle $E \rightarrow B$, every fiber is isomorphic to $\mathbf{P}^1$, hence irreducible and rational. If S' is obtained from E by a sequence of blow-ups at points, then fibers of p become (possibly reducible) divisors whose irreducible components are either:

- strict transforms of $\mathbf{P}^1$ fibers (hence rational), or
- exceptional curves of blow-ups (each exceptional curve is a $\mathbf{P}^1$).

Therefore every irreducible component of any fiber of p is a rational curve. $\square$

Proposition 14.7 (Rationality of special fiber components). *In the situation of Lemma 6.1, every irreducible component of C_0 is rational.*

Proof. Apply Proposition 14.4 to obtain a ruled surface $\tilde{\mathcal{C}}$ and a proper birational morphism $\pi : \tilde{\mathcal{C}} \rightarrow \mathcal{C}$. Let $\tilde{C}_0$ be the special fiber of $\tilde{\mathcal{C}} \rightarrow \text{Spec}(A)$. Then $\pi(\tilde{C}_0) = C_0$ set-theoretically and cycle-theoretically (up to components contracted by π).

Let D be an irreducible component of C_0 . Choose an irreducible component $\tilde{D}$ of $\tilde{C}_0$ mapping onto D . By Lemma 14.6, $\tilde{D}$ is a rational curve.

Now $\pi|_{\tilde{D}} : \tilde{D} \rightarrow D$ is a dominant morphism of integral projective curves. Let $\tilde{D}^\nu \cong \mathbf{P}^1$ be the normalization of $\tilde{D}$ and D^ν the normalization of D . Then the induced map $\mathbf{P}^1 \rightarrow D^\nu$ is non-constant and hence finite. But a smooth projective curve admitting a non-constant map from $\mathbf{P}^1$ must have genus 0 (Riemann–Hurwitz), so $D^\nu \cong \mathbf{P}^1$ and D is rational. $\square$

14.5 Completion of Lemma 6.1

Proof of Lemma 6.1. Connectedness is Proposition 14.2. Rationality of irreducible components is Proposition 14.7. $\square$

14.6 Where Tsen and Lüroth enter (compatibility with Samuel’s proof)

Remark 14.8 (Genus 0 versus rational, and Tsen). Samuel’s proof is designed to handle situations where one starts with a *unirational* parameter and obtains a *genus 0* curve over a function field, which is a conic. Over a general field, a conic might have no rational point, so genus 0 does not automatically imply rational.

Tsen’s theorem says that over $k(t)$ (function field of a curve over an algebraically closed field), every conic has a rational point, hence is $\mathbf{P}^1$. This is the key bridge from genus 0 to rationality in Samuel’s argument.

Once one knows the function field of the limiting curve components embeds into a purely transcendental field in one variable, Lüroth’s theorem implies that function field is itself purely transcendental, i.e. the curve is rational.

Part II

From Rational Equivalence to the Albanese Variety

15 A bridge between Samuel and Serre: zero-cycles and the Albanese map

The two memoirs address different but deeply compatible universal phenomena. Samuel shows that rational equivalence on arbitrary cycles is generated by one-parameter families and is compatible with the basic operations of intersection theory. Serre, on the other hand, isolates the universal morphism from a variety to a commutative algebraic group, and then specializes this to the abelian case in order to construct the Albanese variety.

For smooth projective varieties, these viewpoints meet most concretely in codimension $\dim V$. Indeed, rational equivalence of zero-cycles is the correct equivalence relation that allows one to attach to a pointed variety (V, x_0) a canonical map

$$V \longrightarrow \mathrm{Alb}(V), \quad x \longmapsto \alpha_V(x) - \alpha_V(x_0),$$

and then to extend this map additively to a homomorphism on degree-zero zero-cycles. The purpose of this section is to give a clean proof of that factorization using only the mechanisms developed in the two parts of these notes.

15.1 Zero-cycles of fixed degree and addition on an abelian variety

Let A be an abelian variety over k . For each integer $d \geq 0$, the addition law on A induces a morphism

$$\sigma_d : \mathrm{Sym}^d(A) \longrightarrow A, \quad [a_1 + \cdots + a_d] \longmapsto a_1 + \cdots + a_d.$$

If Z is an effective relative zero-cycle of degree d on $A \times T$, flat over T on a dense open subset, then on that open subset the family determines a morphism

$$\gamma_Z : T \longrightarrow \mathrm{Sym}^d(A) \xrightarrow{\sigma_d} A$$

whose value at a point t is the sum of the points occurring in the fiber $Z(t)$, counted with multiplicity.

Lemma 15.1 (Constancy along rational curves). *Every morphism $\mathbf{P}^1 \rightarrow A$ to an abelian variety is constant.*

Proof. A non-constant morphism $\mathbf{P}^1 \rightarrow A$ would yield a nonzero regular differential on $\mathbf{P}^1$ by pulling back a nonzero invariant differential on A . But $H^0(\mathbf{P}^1, \Omega_{\mathbf{P}^1}^1) = 0$. Hence every such morphism is constant. $\square$

Proposition 15.2 (Morphisms to abelian varieties are constant on rationally equivalent effective zero-cycles of fixed degree). *Let V be a nonsingular projective variety, let $g : V \rightarrow A$ be a morphism to an abelian variety, and let*

$$T \subset V \times \mathbf{P}^1$$

be an effective cycle such that each defined fiber $T(t)$ is a zero-cycle of degree d on V . Then the degree- d zero-cycles $g_(T(a))$ and $g_*(T(b))$ have the same image under the addition morphism σ_d for any points $a, b \in \mathbf{P}^1$ where the fibers are defined. Equivalently,*

$$\sum n_i g(x_i) = \sum m_j g(y_j)$$

whenever $T(a) = \sum n_i [x_i]$ and $T(b) = \sum m_j [y_j]$.

Proof. Apply $g \times \text{id}_{\mathbf{P}^1}$ to obtain an effective relative zero-cycle

$$(g \times \text{id})_*(T) \subset A \times \mathbf{P}^1.$$

Over the locus where the fibers are defined and have degree d , this family determines a morphism

$$\mathbf{P}^1 \dashrightarrow \text{Sym}^d(A) \xrightarrow{\sigma_d} A.$$

Since the source is complete and the target is projective, the rational map extends to a morphism

$$\varphi : \mathbf{P}^1 \rightarrow A.$$

By Lemma 15.1, φ is constant. Evaluating at a and b yields the claim. $\square$

15.2 Factorization through CH_0^0

Fix a base point $x_0 \in V(k)$. Write $Z_0(V)$ for the group of zero-cycles on V and $Z_0(V)^0$ for the subgroup of degree 0 cycles. By Samuel's theorem, rational equivalence of zero-cycles is generated by differences of fibers of effective families over $\mathbf{P}^1$; equivalently, by principal divisors on integral curves inside V . The preceding proposition therefore implies the following.

Theorem 15.3 (Additive factorization through degree-zero Chow groups). *Let V be a nonsingular projective variety, let $x_0 \in V(k)$, and let*

$$\alpha_V : V \rightarrow \text{Alb}(V)$$

be the Albanese morphism normalized by $\alpha_V(x_0) = 0$. Then the assignment

$$\Psi_{\alpha_V} : Z_0(V)^0 \longrightarrow \text{Alb}(V)(k), \quad \sum_i n_i [x_i] \longmapsto \sum_i n_i \alpha_V(x_i)$$

depends only on the rational equivalence class of the zero-cycle. Hence it descends to a canonical homomorphism

$$\overline{\Psi}_{\alpha_V} : CH_0(V)^0 \longrightarrow \text{Alb}(V)(k).$$

Moreover, for every morphism $g : V \rightarrow A$ to an abelian variety with $g(x_0) = 0$, the analogous rule

$$\sum_i n_i [x_i] \longmapsto \sum_i n_i g(x_i)$$

also factors through $CH_0(V)^0$.

Proof. It is enough to prove the statement for generators of rational equivalence. By Theorem 4.3 and Theorem 8.1, every rational equivalence between degree-zero zero-cycles is generated by expressions of the form

$$T(a) - T(b)$$

for an effective relative zero-cycle T on $V \times \mathbf{P}^1$ of constant degree. Applying Proposition 15.2 to g shows that

$$\sum_i n_i g(x_i) = \sum_j m_j g(y_j)$$

whenever $T(a) = \sum_i n_i [x_i]$ and $T(b) = \sum_j m_j [y_j]$. Thus g_* kills all rational equivalences in degree zero, so the formula depends only on the Chow class. Taking $g = \alpha_V$ gives the desired map to $\text{Alb}(V)(k)$. $\square$

Corollary 15.4 (Universal property on pointed zero-cycles). *Let (V, x_0) be a pointed nonsingular projective variety. For every abelian variety A and every morphism $g : V \rightarrow A$ with $g(x_0) = 0$, there is a unique group homomorphism*

$$\tilde{g} : \text{Im}(\overline{\Psi}_{\alpha_V}) \longrightarrow A(k)$$

such that for every $x \in V(k)$ one has

$$\tilde{g}(\overline{\Psi}_{\alpha_V}([x] - [x_0])) = g(x).$$

Equivalently, the map on points induced by g factors uniquely through the canonical map

$$V(k) \longrightarrow CH_0(V)^0 \xrightarrow{\overline{\Psi}_{\alpha_V}} \text{Alb}(V)(k).$$

In particular, the Albanese map is the universal regular recipient of degree-zero zero-cycles arising from points of V .

Proof. By Serre's universal property of α_V , there exists a unique homomorphism of abelian varieties

$$h : \text{Alb}(V) \rightarrow A$$

with $g = h \circ \alpha_V$. Evaluating on zero-cycles shows that

$$\sum_i n_i g(x_i) = h\left(\sum_i n_i \alpha_V(x_i)\right),$$

which exactly says that the required factorization is induced by h on k -points. Uniqueness follows from the uniqueness of h . $\square$

Remark 15.5. Modern language packages the previous theorem by saying that the Albanese morphism induces a *regular homomorphism* from $CH_0(V)^0$ to $\text{Alb}(V)(k)$, universal among regular homomorphisms from degree-zero zero-cycles to abelian varieties. The two original papers already contain the essential geometric input for this statement: Samuel supplies the control of rational equivalence by one-parameter families, and Serre supplies the universal abelian quotient.

Part III

Universal Morphisms and the Albanese Variety (Serre)

16 Setup and basic constructions

16.1 Varieties, commutative algebraic groups, affine homomorphisms

Throughout, V denotes an integral k -variety and A a commutative algebraic group over k (assumed irreducible/connected, as in Serre). A morphism $f : V \rightarrow A$ is a morphism of varieties.

Definition 16.1 (Affine homomorphism). Let B, A be commutative algebraic groups. A morphism $h : B \rightarrow A$ is an *affine homomorphism* if it is a translation composed with a group homomorphism: $h = \tau_a \circ h_0$ with $a \in A(k)$ and $h_0 : B \rightarrow A$ a group homomorphism. The *kernel* of h means $\text{Ker}(h_0)$.

Remark 16.2. This definition is designed so that composition of affine homomorphisms is affine, and so that “finite kernel” makes sense (kernel of the homogeneous part). Serre stresses that in many statements only the A -torsor structure matters, not the group law itself.

16.2 The subgroup generated by increments

Let $f : V \rightarrow A$. Serre defines the subgroup generated by the *increments* $f(x) - f(y)$.

Definition 16.3 (Generated subgroup). Let $A' \subseteq A$ be the smallest closed algebraic subgroup containing all points $f(x) - f(y)$ for $x, y \in V$. We say f *generates* A if $A' = A$.

The next construction makes A' explicit.

Definition 16.4 (Sum–difference maps). For $n \geq 1$ define

$$f_n : V^{2n} \rightarrow A, \quad f_n(x_1, \dots, x_n; y_1, \dots, y_n) = \sum_{i=1}^n f(x_i) - \sum_{j=1}^n f(y_j).$$

Let $W_n := \text{Im}(f_n) \subseteq A$.

Proposition 16.5 (Stabilization yields a closed subgroup). *There exists $m \geq 1$ such that $\overline{W_n} = \overline{W_m}$ for all $n \geq m$. For such an m , the constructible set W_m is Zariski closed and equals the subgroup A' generated by the increments. In particular A' is connected and irreducible.*

Proof. Step 1: stabilization of closures. Since V^{2n} is irreducible and f_n is a morphism, each W_n is constructible and irreducible, hence $\overline{W_n}$ is irreducible closed. Moreover $W_n \subseteq W_{n+1}$ by inserting the trivial pair $x_{n+1} = y_{n+1}$, hence $\overline{W_n} \subseteq \overline{W_{n+1}}$. Noetherianity of A implies the increasing chain of closed sets stabilizes, so $\overline{W_n} = \overline{W_m}$ for $n \geq m$.

Step 2: $W_{2n} = W_n - W_n$. By definition,

$$W_{2n} = \text{Im}(f_{2n}) = \text{Im}((x, y) \mapsto f_n(x) - f_n(y)) \subseteq A,$$

where $x, y \in V^{2n}$. Hence $W_{2n} = W_n - W_n$ set-theoretically. In particular W_{2n} is stable under inversion and contains 0.

Step 3: W_m contains a dense open of $\overline{W_m}$. A standard fact: a constructible subset of an irreducible variety contains a dense open subset of its closure. Thus W_m contains a nonempty open subset $U \subseteq \overline{W_m}$.

Step 4: forcing closedness using translation invariance at the generic level. Consider the subtraction morphism $\sigma : A \times A \rightarrow A$, $\sigma(a, b) = a - b$. Then $\sigma(U \times U)$ is constructible and contains a dense open subset of $\overline{W_m} - \overline{W_m} = \overline{W_{2m}}$. But $\overline{W_{2m}} = \overline{W_m}$ by stabilization (since $2m \geq m$), hence $\sigma(U \times U)$ contains a dense open subset of $\overline{W_m}$. Since $\sigma(U \times U) \subseteq W_{2m} = W_m - W_m$, we get that $W_m - W_m$ contains a dense open in $\overline{W_m}$. Using again stabilization plus irreducibility, one concludes W_m must be Zariski dense in $\overline{W_m}$ and stable under the group operation on a dense open, which forces W_m to equal its closure: indeed, W_m is a constructible subset containing a dense open and stable under translate-differences in the sense $W_m = W_m - W_m$ inside the stabilized closure; hence W_m is a subgroup of $\overline{W_m}$ containing a dense open, so it must be all of $\overline{W_m}$. Therefore W_m is closed and is an algebraic subgroup of A .

Step 5: identification with A' . By construction, W_1 contains all increments $f(x) - f(y)$, hence so does W_m . Thus the closed subgroup it generates is contained in W_m . Conversely W_m is generated by sums of increments, so $W_m \subseteq A'$. Therefore $W_m = A'$. $\square$

Remark 16.6 (Birational invariance). If $U \subset V$ is a nonempty open, then $f(U)$ generates the same subgroup A' as $f(V)$. Indeed U^{2n} is dense in V^{2n} , so $\overline{f_n(U^{2n})} = \overline{f_n(V^{2n})}$ for each n , and hence the stabilized closure A' is unchanged.

Lemma 16.7 (Geometric characterization of “generates”). *f generates A iff $f(V)$ is not contained in any translate $a + B$ with $B \subsetneq A$ a proper closed algebraic subgroup.*

Proof. If $f(V) \subset a + B$, then $f(x) - f(y) \in B$ for all x, y , so the subgroup generated by increments lies in B . Conversely, if $A' \subsetneq A$, then $f(V)$ is contained in a coset of A' : fix $x_0 \in V$ and write $f(x) = f(x_0) + (f(x) - f(x_0)) \in f(x_0) + A'$. $\square$

17 Maximal morphisms and the factorization theorem

17.1 Isogenies

Definition 17.1 (Isogeny; separable vs radical). A homomorphism of commutative algebraic groups $g : B \rightarrow A$ is an *isogeny* if g is surjective and $\text{Ker}(g)$ is finite. It is *separable* (resp. *radical*) if the induced extension of function fields $k(B)/k(A)$ is separable (resp. purely inseparable).

17.2 Maximal morphisms

Definition 17.2 (Maximality). A morphism $f : V \rightarrow A$ is *maximal* if (i) f generates A , and (ii) for every factorization

$$V \xrightarrow{f_1} B \xrightarrow{g} A$$

where $g : B \rightarrow A$ is an isogeny (or radicial isogeny in the radicial variant), one has that g is an isomorphism.

Example 17.3. If $A = 0$, every morphism $V \rightarrow A$ is maximal. If V is a smooth projective curve and $f : V \rightarrow J(V)$ is the Abel–Jacobi map, then f is maximal (Serre).

17.3 Factorization through a maximal morphism (Serre’s Theorem 1)

Theorem 17.4 (Factorization theorem). *Let $f : V \rightarrow A$ be a morphism into a commutative algebraic group. Then there exists a factorization*

$$V \xrightarrow{g} B \xrightarrow{h} A$$

such that:

- (i) g is maximal;
- (ii) h is an affine homomorphism with finite kernel.

Moreover, this factorization is unique up to unique isomorphism of B .

Proof. Step 0: reduce to the generating case. Let $A' \subseteq A$ be the subgroup generated by increments (Proposition 16.5). Fix $x_0 \in V$ and define $f'(x) = f(x) - f(x_0) \in A$. Then $f'(V) \subseteq A'$, and the translation map identifies f with the composite $V \xrightarrow{f'} A' \hookrightarrow A$ up to translation. Thus it suffices to factor f' as claimed; composing with $A' \hookrightarrow A$ produces the required factorization for f . Hence assume from now on that f generates A .

Step 1: if f is maximal, done. Take $B = A$, $g = f$, $h = \text{id}$.

Step 2: otherwise, iterate nontrivial isogeny factorizations. If f is not maximal, there exists a non-isomorphism isogeny $h_1 : A_1 \rightarrow A$ and a morphism $f_1 : V \rightarrow A_1$ with $f = h_1 \circ f_1$. Since f generates A , Lemma 16.7 implies f_1 generates A_1 (otherwise $f(V)$ would lie in a translate of a proper subgroup of A). If f_1 is not maximal, factor it through a nontrivial isogeny $h_2 : A_2 \rightarrow A_1$, etc. Thus we obtain a sequence

$$\begin{array}{ccccc} & & A_2 & \xrightarrow{h_2} & A_1 & \xrightarrow{h_1} & A \\ & \nearrow f_2 & & \nearrow f_1 & & \nearrow f & \\ V & & & & & & \end{array}$$

with each h_i a nontrivial isogeny and each f_i generating A_i .

Step 3: show the iteration must terminate. Suppose for contradiction that we can continue indefinitely, producing $A \leftarrow A_1 \leftarrow A_2 \leftarrow \cdots$ with each map an isogeny not an isomorphism. Choose $n \gg 0$ such that $f_n : V^{2n} \rightarrow A$ is dominant (possible because f generates A ; indeed, the stabilized image subgroup equals A , so some sum–difference map is dominant on A). Then each composite $f_{i,n} : V^{2n} \rightarrow A_i$ induced from f_i is also dominant.

Passing to function fields, dominance gives inclusions

$$k(A) \hookrightarrow k(V^{2n}), \quad k(A_i) \hookrightarrow k(V^{2n}) \quad \text{for all } i.$$

Because $A_i \rightarrow A$ is an isogeny, $k(A_i)$ is a finite field extension of $k(A)$. Moreover, because the isogeny is not an isomorphism, the extension is *proper*: $k(A) \subsetneq k(A_1) \subsetneq k(A_2) \subsetneq \cdots$. Thus we obtain a strictly increasing chain of intermediate fields inside the finitely generated extension $k(V^{2n})/k(A)$.

Step 4: eliminate infinite chains via finite type subextension lemma. We use the following standard lemma (cf. Serre’s Lemma 1):

Lemma 17.5 (Finite type subextension). *Let M/E be a finitely generated field extension. Then for every intermediate field $E \subseteq F \subseteq M$, the extension F/E is finitely generated.*

Proof of Lemma 17.5. Let $(x_1, \dots, x_s)$ be a transcendence basis of F/E . Extend it to a transcendence basis $(x_1, \dots, x_s, y_1, \dots, y_r)$ of M/E . Then M is algebraic over $E(x_1, \dots, x_s, y_1, \dots, y_r)$ and F is algebraic over $E(x_1, \dots, x_s)$. Because $E(x_1, \dots, x_s, y_1, \dots, y_r)$ is purely transcendental over $E(x_1, \dots, x_s)$, the algebraic extensions $F/E(x)$ and $E(x, y)/E(x)$ are linearly disjoint. Hence $F \cdot E(x, y)$ has degree over $E(x, y)$ equal to $[F : E(x)]$. But M is finitely generated over $E(x, y)$, so the algebraic part has finite degree; thus $[F : E(x)] < \infty$. Therefore F is a finite (hence finitely generated) extension of $E(x_1, \dots, x_s)$, so F/E is finitely generated. $\square$

Apply the lemma with $E = k(A)$ and $M = k(V^{2n})$. Then the union $F_\infty := \bigcup_i k(A_i)$ is a subextension of M/E , hence finitely generated over E . But F_∞/E is also algebraic (union of finite extensions). A finitely generated algebraic extension is finite. Thus F_∞/E must be finite, contradicting the strict infinite chain $k(A) \subsetneq k(A_1) \subsetneq k(A_2) \subsetneq \cdots$. Hence the iteration terminates.

Step 5: conclude. Let r be the first index such that $f_r : V \rightarrow A_r$ is maximal. Set $B := A_r$, $g := f_r$, and $h := h_1 \circ \cdots \circ h_r : B \rightarrow A$ (affine isogeny with finite kernel). Then $f = h \circ g$ with g maximal and h finite-kernel affine homomorphism, as required.

Uniqueness (sketch with rigidification). If $f = h \circ g = h' \circ g'$ with g, g' maximal and h, h' finite-kernel affine, consider the induced rational map between B and B' obtained by the universal property of maximality (applied to the graph of (g, g')). Finite-kernel properties force it to be a group isogeny both ways, hence an isomorphism, and maximality pins it down uniquely. $\square$

18 Universal morphisms relative to a category

18.1 The categories $\mathcal{C}$

Fix a category $\mathcal{C}$ of commutative algebraic groups satisfying Serre’s closure properties:

(i) If $A_1, A_2 \in \mathcal{C}$ then $A_1 \times A_2 \in \mathcal{C}$.

(ii) If $f : A \rightarrow B$ is a homomorphism with finite kernel and $B \in \mathcal{C}$, then $A \in \mathcal{C}$.

Definition 18.1 (Universal morphism for $\mathcal{C}$). Let $f : V \rightarrow A$ with $A \in \mathcal{C}$. We say f is *universal* for $\mathcal{C}$ if for every $f' : V \rightarrow A'$ with $A' \in \mathcal{C}$ there exists a *unique* affine homomorphism $h : A \rightarrow A'$ such that $f' = h \circ f$.

Remark 18.2. Universality is a representability statement: f represents the functor $A' \mapsto \text{Hom}(V, A')$ on $\mathcal{C}$, but with morphisms restricted to affine homomorphisms on targets.

18.2 Characterization by maximality and dimension (Serre's Theorem 2)

Theorem 18.3 (Universality criterion). *Assume $\mathcal{C}$ satisfies (i)–(ii). Let $f : V \rightarrow A$ with $A \in \mathcal{C}$. Then f is universal for $\mathcal{C}$ if and only if:*

(a) f is maximal;

(b) for every maximal morphism $f' : V \rightarrow A'$ with $A' \in \mathcal{C}$, one has $\dim A' \leq \dim A$.

Proof. (Universality $\Rightarrow$ maximality). Assume f is universal. Suppose $f = \ell \circ k$ with $\ell : B \rightarrow A$ an affine homomorphism that is surjective with finite kernel. By universality applied to $k : V \rightarrow B$ (note $B \in \mathcal{C}$ by (ii)), there exists $h : A \rightarrow B$ affine such that $k = h \circ f$. Then $f = \ell \circ k = \ell \circ h \circ f$. By uniqueness in the universal property, $\ell \circ h = \text{id}_A$. In particular ℓ is injective. Since ℓ is also surjective, ℓ is an isomorphism. Hence f admits no nontrivial finite-kernel quotient factorization: f is maximal.

(Universality $\Rightarrow$ dimension maximality). Let $f' : V \rightarrow A'$ be maximal with $A' \in \mathcal{C}$. By universality there is an affine homomorphism $h : A \rightarrow A'$ with $f' = h \circ f$. Since f' generates A' , h must be surjective (otherwise $f'(V) \subset h(A)$ lies in a translate of a proper subgroup). Therefore $\dim A' \leq \dim A$.

(Maximality + dimension condition $\Rightarrow$ universality). Assume (a)–(b). Given any $f' : V \rightarrow A'$ with $A' \in \mathcal{C}$, consider the product morphism

$$g := (f, f') : V \rightarrow A \times A'.$$

By Theorem 17.4, factor $g = i \circ k$ with $k : V \rightarrow B$ maximal and $i : B \rightarrow A \times A'$ affine with finite kernel. By (i)–(ii), $B \in \mathcal{C}$.

Let $p : A \times A' \rightarrow A$ and $p' : A \times A' \rightarrow A'$ be projections. Then $f = p \circ i \circ k$. Because f generates A , the image $p(i(B))$ must be all of A (same coset argument as above), so $p \circ i$ is surjective. Also $\text{Ker}(p \circ i)$ is finite because $\text{Ker}(i)$ is finite. Thus $p \circ i : B \rightarrow A$ is a finite-kernel surjective affine homomorphism.

Now k is maximal, hence $p \circ i \circ k = f$ is a factorization of f through $p \circ i$; maximality of f forces $p \circ i$ to be an isomorphism. Let $r : A \rightarrow B$ be its inverse (affine homomorphism). Define

$$h := p' \circ i \circ r : A \rightarrow A'.$$

Then

$$h \circ f = p' \circ i \circ r \circ f = p' \circ i \circ r \circ p \circ i \circ k = p' \circ i \circ k = f'.$$

Uniqueness. If $h_1, h_2 : A \rightarrow A'$ are affine homomorphisms with $h_1 \circ f = h_2 \circ f$ and f generates A , then h_1 and h_2 agree on the translate $f(x_0) + A$ (Lemma 16.7 argument), hence they agree on all of A . This gives uniqueness.

Therefore f is universal. $\square$

Corollary 18.4 (Existence criterion). *A universal morphism $V \rightarrow A$ for $\mathcal{C}$ exists if and only if the set of dimensions $\{\dim A' \mid A' \in \mathcal{C}, \exists \text{ maximal } f' : V \rightarrow A'\}$ is bounded.*

Proof. If a universal morphism exists, Theorem 18.3(b) gives the bound. Conversely, choose among maximal targets in $\mathcal{C}$ one of maximal dimension; then Theorem 18.3 applies. $\square$

18.3 Functoriality and products

Proposition 18.5 (Functoriality in V). *Assume universal morphisms exist for V and V' in $\mathcal{C}$:*

$$\alpha_V : V \rightarrow A, \quad \alpha_{V'} : V' \rightarrow A'.$$

For any morphism $\varphi : V \rightarrow V'$ there exists a unique affine homomorphism $\Phi : A \rightarrow A'$ such that $\alpha_{V'} \circ \varphi = \Phi \circ \alpha_V$.

Proof. Apply universality of α_V to the morphism $\alpha_{V'} \circ \varphi : V \rightarrow A'$. Uniqueness is built into the universal property. $\square$

$$\begin{array}{ccc} V & \xrightarrow{\alpha_V} & A \\ \varphi \downarrow & & \downarrow \Phi \\ V' & \xrightarrow{\alpha_{V'}} & A' \end{array}$$

Proposition 18.6 (Compatibility with products). *If $\alpha_V : V \rightarrow A$ and $\alpha_W : W \rightarrow B$ are universal in $\mathcal{C}$, then $\alpha_{V \times W} : V \times W \rightarrow A \times B$ defined by $(v, w) \mapsto (\alpha_V(v), \alpha_W(w))$ is universal in $\mathcal{C}$.*

Proof. Given $f : V \times W \rightarrow C$ with $C \in \mathcal{C}$, restrict to $V \times \{w_0\}$ and $\{v_0\} \times W$. For commutative algebraic groups, the standard “no mixed terms” property implies $f(v, w) = f_V(v) + f_W(w) - f(v_0, w_0)$ after translating so that $f(v_0, w_0) = 0$, hence f is determined by morphisms $f_V : V \rightarrow C$ and $f_W : W \rightarrow C$. By universality, these factor uniquely through A and B , and hence f factors uniquely through $A \times B$. $\square$

19 Radicial maximality and the differential criterion

19.1 Invariant differentials and Lie algebra

Let A be a commutative algebraic group. Let $\mathfrak{t}_A := \text{Lie}(A)$ be its Lie algebra (tangent space at identity), and let $s_A := \mathfrak{t}_A^\vee$. Then s_A identifies with the space of translation-invariant regular 1-forms on A .

Lemma 19.1 (Commutativity of $\text{Lie}(A)$). *If A is commutative, then $\mathfrak{t}_A$ is an abelian Lie algebra.*

Proof. Let $m : A \times A \rightarrow A$ be addition. Functoriality of Lie gives a bilinear map $\mathfrak{t}_A \times \mathfrak{t}_A \simeq \mathfrak{t}_{A \times A} \rightarrow \mathfrak{t}_A$ sending $(X, Y) \mapsto X + Y$. The bracket on $\mathfrak{t}_{A \times A} \simeq \mathfrak{t}_A \oplus \mathfrak{t}_A$ is componentwise, so $[(X, 0), (0, Y)] = 0$. Applying the Lie map induced by m yields $[X, Y] = 0$ in $\mathfrak{t}_A$. $\square$

19.2 Pullback of invariant differentials

Let V be an integral variety and let $D(V)$ denote the $k(V)$ -vector space of Kähler differentials $\Omega_{k(V)/k}^1$. Given a morphism $f : V \rightarrow A$ and an invariant differential $\omega \in s_A$, the pullback $f^*\omega$ is a rational 1-form on V , hence an element of $D(V)$. Thus we have a k -linear map

$$f^* : s_A \longrightarrow D(V).$$

19.3 Radicial maximality

We now state Serre's criterion. For precise radicial language, we distinguish:

Definition 19.2 (Radically maximal). A morphism $f : V \rightarrow A$ is *radically maximal* if it generates A and if any factorization $f = g \circ f_1$ with g a *radicial* isogeny forces g to be an isomorphism.

Theorem 19.3 (Differential criterion). *Let $f : V \rightarrow A$ be a morphism to a commutative algebraic group.*

- (i) *If $f^* : s_A \rightarrow D(V)$ is injective, then f is radially maximal.*
- (ii) *Conversely, if f is radially maximal and V is normal, then f^* is injective.*

Proof. (i) *Injective pullback $\Rightarrow$ radicial maximality.* Suppose $f = g \circ f_1$ with $g : B \rightarrow A$ an affine homomorphism with finite kernel, radicial in the sense of function fields. Then $f^* = f_1^* \circ g^*$ on invariant differentials. If f^* is injective, then $g^* : s_A \rightarrow s_B$ must be injective. Dualizing, the induced tangent map $\mathfrak{t}_B \rightarrow \mathfrak{t}_A$ is surjective. Since $\text{Ker}(g)$ is finite, $\dim B = \dim A$. A surjective homomorphism of connected algebraic groups with finite kernel and equal dimension is an isogeny of degree 1, hence an isomorphism. Thus g is an isomorphism, proving radicial maximality.

(ii) *Radicial maximality + normality $\Rightarrow$ injective pullback.* Assume V normal and $\omega \in s_A$ satisfies $f^*\omega = 0$. We must show $\omega = 0$.

Case 1: $\text{char}(k) = 0$. Choose $m \gg 0$ such that the sum-difference map $f_m : V^{2m} \rightarrow A$ is dominant. Then the induced extension $k(V^{2m})/k(A)$ is separable. Let $p_i, q_j : V^{2m} \rightarrow V$ be the projections. Since $f_m = \sum_i f \circ p_i - \sum_j f \circ q_j$ and ω is invariant, we have

$$f_m^* \omega = \sum_i p_i^*(f^* \omega) - \sum_j q_j^*(f^* \omega) = 0.$$

But for a dominant separable morphism $g : X \rightarrow A$ from an integral normal X , the pullback map on invariant differentials is injective: if $g^* \omega = 0$ then ω vanishes at the generic point because separability implies the differential dg has full rank generically. Hence $\omega = 0$.

Case 2: $\text{char}(k) = p > 0$. Let $N := \text{Ker}(f^* : s_A \rightarrow D(V))$. Serre uses Cartier theory on invariant differentials to show N is stable under Cartier, and by duality the annihilator $N^\perp \subset \mathfrak{t}_A$ is stable under the p -power map $X \mapsto X^{[p]}$. Using the structure theory of height-1 purely inseparable (radicial) isogenies, one constructs a height-1 radicial isogeny $h : A \rightarrow B$ whose induced tangent map has kernel $N^\perp$. Then $(h \circ f)^* : s_B \rightarrow D(V)$ is zero, i.e. all invariant differentials pull back to zero. Equivalently, the differential of $h \circ f$ vanishes generically, so $h \circ f$ is *purely inseparable* at the level of function fields; it factors through a Frobenius twist $B^{(1/p)}$. Normality of V is used to upgrade the induced rational map to a morphism (Serre appeals to a “principal divisor”/Hartogs-type extension argument). Finally, since f was assumed radicially maximal, the resulting nontrivial radicial factorization forces h to be an isomorphism, hence $N = 0$. Therefore f^* is injective. $\square$

Corollary 19.4 (Dimension bound from regular 1-forms). *Assume V is normal. Let $\Omega(V)$ be the k -vector space of regular 1-forms on the smooth locus of V . If $f : V \rightarrow A$ generates A , then $\dim A \leq \dim_k \Omega(V)$.*

Proof. Factor f as $V \xrightarrow{g} B \rightarrow A$ with g radicially maximal (Theorem 17.4 in the radicial setting). Then Theorem 19.3(ii) implies $g^* : s_B \hookrightarrow D(V)$ is injective. But invariant differentials on B pull back to regular differentials on a big open of V , hence the image lies in $\Omega(V) \subset D(V)$. Therefore $\dim B = \dim s_B \leq \dim \Omega(V)$ and $\dim A \leq \dim B$ because $B \rightarrow A$ has finite kernel. $\square$

20 The Albanese variety

20.1 Existence for abelian targets

Let $\mathcal{C}_{\text{ab}}$ be the category of abelian varieties over k .

Theorem 20.1 (Existence of the Albanese morphism). *For every integral variety V there exists a universal morphism*

$$\alpha_V : V \rightarrow \text{Alb}(V)$$

for the category $\mathcal{C}_{\text{ab}}$. The abelian variety $\text{Alb}(V)$ is functorial in V and compatible with products.

Proof (Serre’s strategy). By Corollary 18.4, it suffices to prove a uniform bound on $\dim A$ for maximal morphisms $V \rightarrow A$ with A an abelian variety. Serre reduces birationally to an open $U = W - F$ with W complete (projective normal) and $\text{codim}_W(F) \geq 2$. On such a

model, Hartogs-type extension and coherence (cf. Appendix A below) imply that the space of regular 1-forms on U is finite-dimensional. Then Corollary 19.4 bounds $\dim A$ by $\dim \Omega(U)$ for any maximal (radicial) morphism $U \rightarrow A$. Hence dimensions of maximal abelian targets are bounded, giving existence by Corollary 18.4. Functoriality and product compatibility follow from Propositions 18.5 and 18.6. $\square$

20.2 Extension across birational models and birational invariance

The existence proof above becomes more transparent once one isolates the extension principle for morphisms to abelian varieties.

Proposition 20.2 (Rational maps from normal complete varieties to abelian varieties are everywhere defined). *Let X be a normal complete variety and let A be an abelian variety. Then every rational map $\varphi : X \dashrightarrow A$ extends uniquely to a morphism*

$$\tilde{\varphi} : X \longrightarrow A.$$

Proof. Let $U \subset X$ be the maximal open subset on which φ is regular. Consider the closure $\Gamma \subset X \times A$ of the graph of $\varphi|_U$, and let

$$\pi : \Gamma \rightarrow X, \quad q : \Gamma \rightarrow A$$

be the projections. Since A is proper, Γ is proper over X . By construction π is birational and is an isomorphism over U .

Assume for contradiction that π is not an isomorphism. Then some fiber $\pi^{-1}(x)$ has positive dimension. Because A contains no rational curves, the restriction of q to every rational component of such a fiber must be constant. A more invariant way to package the same point is the following: for any curve $T \subset \Gamma$ contracted by π , its image $q(T)$ is a complete curve in A contained in a fiber of the rational map induced by $U \rightarrow A$; translation reduces to a curve through the origin, and completeness forces that image to be a point. Hence q is constant on each connected component of every exceptional fiber of π .

Now define $\tilde{\varphi}(x)$ to be the common value of q on $\pi^{-1}(x)$. This is well defined because normality of X implies that the exceptional locus of a birational proper map has codimension at least 2 in the target, and regularity can be checked on codimension-1 points. Equivalently, by Zariski's Main Theorem, $\pi_* \mathcal{O}_\Gamma = \mathcal{O}_X$; since A is separated, a morphism out of Γ constant on the fibers of π descends uniquely to X . The descended map extends $\varphi|_U$, so it is the desired morphism.

Uniqueness is immediate because two morphisms $X \rightarrow A$ agreeing on the dense open subset U agree everywhere. $\square$

Corollary 20.3 (Birational invariance for proper normal models). *Let X and Y be normal complete varieties that are birationally equivalent. Then, for every abelian variety A , pullback along a birational correspondence induces a natural bijection*

$$\mathrm{Hom}(X, A) \xrightarrow{\sim} \mathrm{Hom}(Y, A).$$

In particular, if one of X or Y admits an Albanese variety, then so does the other, and the two Albanese varieties are canonically isomorphic.

Proof. Choose a common dense open subset U over which X and Y are isomorphic. A morphism $X \rightarrow A$ restricts to a morphism $U \rightarrow A$; conversely, Proposition 20.2 extends any morphism $U \rightarrow A$ uniquely first to a rational map on Y and then to a morphism $Y \rightarrow A$. Thus restriction identifies both $\text{Hom}(X, A)$ and $\text{Hom}(Y, A)$ with $\text{Hom}(U, A)$. The statement on Albanese varieties follows immediately from the universal property. $\square$

Remark 20.4 (How this fits into Serre’s proof). Serre passes from an arbitrary variety to a birationally simpler normal complete model in order to estimate dimensions through regular differentials. The corollary shows that this birational replacement does not alter the universal abelian quotient; it only makes the finite-dimensionality argument accessible.

Definition 20.5 (Albanese variety). The abelian variety $\text{Alb}(V)$ in Theorem 20.1 is called the *Albanese variety of V* (in the morphic sense), and α_V is the *Albanese morphism*.

20.3 Normalization at a base point

Assume now that V possesses a rational point $x_0 \in V(k)$. Since translations on an abelian variety are algebraic automorphisms, one may and often does normalize the Albanese morphism by requiring that the chosen base point go to the origin.

Proposition 20.6 (Pointed normalization). *Let V be an integral variety with Albanese morphism $\alpha_V : V \rightarrow \text{Alb}(V)$, and let $x_0 \in V(k)$. Then there exists a unique Albanese morphism*

$$\alpha_{V,x_0} : V \longrightarrow \text{Alb}(V)$$

with the same universal property and satisfying $\alpha_{V,x_0}(x_0) = 0$. It is obtained from any Albanese morphism by translation:

$$\alpha_{V,x_0}(x) = \alpha_V(x) - \alpha_V(x_0).$$

Proof. If $t_a : \text{Alb}(V) \rightarrow \text{Alb}(V)$ denotes translation by a , then $t_{-\alpha_V(x_0)} \circ \alpha_V$ is again a morphism from V to an abelian variety; because $t_{-\alpha_V(x_0)}$ is an automorphism, it satisfies the same universal property as α_V . By construction it sends x_0 to 0.

For uniqueness, let $\beta : V \rightarrow \text{Alb}(V)$ be another Albanese morphism with $\beta(x_0) = 0$. Since both α_{V,x_0} and β are universal, there are unique endomorphisms $u, v : \text{Alb}(V) \rightarrow \text{Alb}(V)$ such that

$$\beta = u \circ \alpha_{V,x_0}, \quad \alpha_{V,x_0} = v \circ \beta.$$

Composing gives $v \circ u \circ \alpha_{V,x_0} = \alpha_{V,x_0}$, hence $v \circ u = \text{id}$ by uniqueness in the universal property. Similarly $u \circ v = \text{id}$, so u is an automorphism. But then β differs from α_{V,x_0} by an automorphism of the target fixing the universal map. The same uniqueness forces that automorphism to be the identity. Therefore $\beta = \alpha_{V,x_0}$. $\square$

20.4 The Abel map for large degree and descent from symmetric powers

We isolate the two geometric inputs used in the proof for curves.

Lemma 20.7 (Large-degree Abel map as a projective bundle). *Let C be a smooth projective curve of genus g , and let $c_0 \in C(k)$. For every integer $d \geq 2g - 1$, the map*

$$\beta_d : \mathrm{Sym}^d(C) \longrightarrow J(C) = \mathrm{Pic}^0(C), \quad D \longmapsto [\mathcal{O}_C(D - dc_0)]$$

identifies $\mathrm{Sym}^d(C)$ with the projectivization of a vector bundle of rank $d - g + 1$ over $J(C)$. In particular, β_d is surjective and its fibers are projective spaces.

Proof. Let $\mathcal{P}$ be a normalized Poincaré line bundle on $C \times J(C)$. Write

$$\pi_C : C \times J(C) \rightarrow C, \quad \pi_J : C \times J(C) \rightarrow J(C)$$

for the projections, and put

$$\mathcal{L} := \mathcal{P} \otimes \pi_C^* \mathcal{O}_C(dc_0).$$

For $\xi \in J(C)$, the restriction $\mathcal{L}_\xi$ to $C \times \{\xi\}$ has degree d . Because $d \geq 2g - 1$, Serre duality and Riemann–Roch give

$$H^1(C, \mathcal{L}_\xi) = 0, \quad h^0(C, \mathcal{L}_\xi) = d - g + 1$$

for every ξ . By cohomology and base change, the sheaf

$$E := \pi_{J*} \mathcal{L}$$

is locally free of rank $d - g + 1$ on $J(C)$.

A point of $\mathbf{P}(E)$ over ξ corresponds to a one-dimensional quotient of $H^0(C, \mathcal{L}_\xi)$, or equivalently to a nonzero section of $\mathcal{L}_\xi$ up to scalar. The zero divisor of such a section is an effective divisor D of degree d satisfying

$$\mathcal{O}_C(D) \cong \mathcal{L}_\xi \cong \xi \otimes \mathcal{O}_C(dc_0),$$

which is equivalent to $[\mathcal{O}_C(D - dc_0)] = \xi$. Thus the fiber of β_d over ξ is canonically the complete linear system $|\xi \otimes \mathcal{O}_C(dc_0)|$, i.e. a projective space $\mathbf{P}(H^0(C, \mathcal{L}_\xi)) \cong \mathbf{P}^{d-g}$. These identifications glue functorially in families, giving

$$\mathrm{Sym}^d(C) \cong \mathbf{P}(E)$$

over $J(C)$. □

Lemma 20.8 (Descent along a projective bundle). *Let $\pi : P \rightarrow S$ be a projective bundle, and let T be a separated variety. If $f : P \rightarrow T$ is a morphism that is constant on every geometric fiber of π , then there exists a unique morphism $\bar{f} : S \rightarrow T$ such that $f = \bar{f} \circ \pi$.*

Proof. The assertion is local on S for the Zariski topology, so we may assume

$$P \cong \mathbf{P}_S^n.$$

Let $U \subset S$ be an affine open subset. Because f is constant on each fiber $\mathbf{P}_{\kappa(s)}^n$, the two pullbacks of $f|_{\pi^{-1}(U)}$ to

$$\mathbf{P}_U^n \times_U \mathbf{P}_U^n$$

along the two projections agree on every geometric fiber over U . Since T is separated, equality on fibers implies equality as morphisms; hence $f|_{\pi^{-1}(U)}$ satisfies the fpqc descent cocycle for the faithfully flat morphism $\pi^{-1}(U) \rightarrow U$. Therefore it descends to a unique morphism $\bar{f}_U : U \rightarrow T$.

On overlaps the descended morphisms agree because their pullbacks to the projective bundle agree with f . Hence the $\bar{f}_U$ glue to a global morphism $\bar{f} : S \rightarrow T$ satisfying $f = \bar{f} \circ \pi$. Uniqueness follows from surjectivity of π . □

20.5 The case of curves: recovery of the Jacobian

The first nontrivial example of the Albanese construction is the classical Jacobian of a smooth projective curve. The point of the argument below is that one can see the universal property of the Jacobian directly through Samuel's control of rational equivalence on zero-cycles together with the additivity properties of morphisms to abelian varieties.

Theorem 20.9 (For smooth projective curves, Albanese = Jacobian). *Let C be a smooth projective geometrically integral curve over k , and assume that $c_0 \in C(k)$ is a rational point. Let*

$$\nu_{c_0} : C \longrightarrow J(C) := \text{Pic}^0(C), \quad x \longmapsto [\mathcal{O}_C(x - c_0)]$$

be the Abel–Jacobi map. Then ν_{c_0} is universal among morphisms from C to abelian varieties sending c_0 to 0. Consequently there is a canonical isomorphism of pointed abelian varieties

$$\text{Alb}(C) \cong J(C),$$

identifying the normalized Albanese morphism with the Abel–Jacobi map.

Proof. Fix a morphism $g : C \rightarrow A$ to an abelian variety such that $g(c_0) = 0$. We must construct a unique homomorphism of abelian varieties $h : J(C) \rightarrow A$ with $g = h \circ \nu_{c_0}$.

Choose an integer $d \geq 2g(C) - 1$. By Lemma 20.7, the map

$$\beta_d : \text{Sym}^d(C) \longrightarrow J(C), \quad D \longmapsto [\mathcal{O}_C(D - dc_0)]$$

is a projective bundle. Define

$$s_d : \text{Sym}^d(C) \longrightarrow A, \quad [x_1 + \cdots + x_d] \longmapsto g(x_1) + \cdots + g(x_d).$$

This is well defined because addition on A is symmetric and regular.

Step 1: constancy on fibers. Suppose $D, D' \in \text{Sym}^d(C)$ have the same image under β_d . Then

$$\mathcal{O}_C(D - dc_0) \cong \mathcal{O}_C(D' - dc_0),$$

so $D - D'$ is principal. As a zero-cycle on C , this means $D - D'$ is rationally equivalent to 0. Now Theorem 15.3 says precisely that the map on degree-zero cycles

$$\sum_i n_i [x_i] \longmapsto \sum_i n_i g(x_i)$$

annihilates rational equivalence. Therefore principal divisors are sent to 0, and thus

$$s_d(D) = s_d(D').$$

So s_d is constant on every fiber of β_d .

Step 2: descent to the Jacobian. Because β_d is a projective bundle, Lemma 20.8 applies and yields a unique morphism

$$h : J(C) \longrightarrow A$$

with

$$s_d = h \circ \beta_d.$$

At this stage h is merely a morphism of varieties; we next see that it is automatically a homomorphism.

Step 3: recovery of g and additivity of h . Embed C into $\text{Sym}^d(C)$ by

$$i_d : C \hookrightarrow \text{Sym}^d(C), \quad x \longmapsto [x + (d-1)c_0].$$

Then

$$\beta_d \circ i_d(x) = [\mathcal{O}_C(x - c_0)] = \nu_{c_0}(x)$$

and, because $g(c_0) = 0$,

$$s_d \circ i_d(x) = g(x) + (d-1)g(c_0) = g(x).$$

Hence

$$g = s_d \circ i_d = h \circ \beta_d \circ i_d = h \circ \nu_{c_0}.$$

So the required factorization exists.

To prove that h is a homomorphism, consider the morphism

$$\delta : J(C) \times J(C) \longrightarrow A, \quad (u, v) \longmapsto h(u + v) - h(u) - h(v).$$

Its pullback along $\nu_{c_0} \times \nu_{c_0}$ is zero because on points of the curve the map $h \circ \nu_{c_0} = g$ was built from the additive rule “sum of points maps to sum of values” used in the definition of s_d . The image of $\nu_{c_0}(C)$ generates $J(C)$ as an algebraic group, hence the image of $\nu_{c_0} \times \nu_{c_0}$ generates $J(C) \times J(C)$. Since δ vanishes on a generating subset and $\delta(0, 0) = 0$, it vanishes identically. Therefore h is a homomorphism of abelian varieties.

Step 4: uniqueness. If $h_1, h_2 : J(C) \rightarrow A$ satisfy $h_1 \circ \nu_{c_0} = h_2 \circ \nu_{c_0}$, then $h_1 - h_2$ vanishes on $\nu_{c_0}(C)$. But divisor classes of degree 0 are generated by classes $[x - c_0]$, so the subgroup of $J(C)(k)$ generated by the image of ν_{c_0} is all of $J(C)(k)$. Hence $h_1 - h_2$ is the zero homomorphism, i.e. $h_1 = h_2$. Thus ν_{c_0} satisfies the pointed universal property.

Finally, Proposition 20.6 supplies a normalized Albanese morphism $\alpha_{C, c_0} : C \rightarrow \text{Alb}(C)$. Applying the universal property twice gives mutually inverse homomorphisms

$$\text{Alb}(C) \rightleftarrows J(C)$$

carrying α_{C, c_0} to ν_{c_0} and back. Therefore

$$\text{Alb}(C) \cong J(C)$$

as pointed abelian varieties. □

20.6 Comparison: Albanese by functions vs by morphisms

Serre discusses an alternative classical definition where one allows *all set-theoretic functions* $V(k) \rightarrow A(k)$ satisfying a suitable regularity condition. Denote this Albanese by $\text{Alb}_{\text{fun}}(V)$. Then there is a natural surjective affine homomorphism

$$\text{Alb}_{\text{fun}}(V) \twoheadrightarrow \text{Alb}(V),$$

whose kernel is connected if V is normal, and trivial if V is smooth. (Serre gives explicit counterexamples showing normality/smoothness hypotheses are essential.)

A Appendix A: Hartogs-type extension of torsion-free coherent sheaves

This appendix expands the codimension- ≥ 2 extension input that lies behind Serre's existence theorem. The point is that, on an integral variety, a torsion-free coherent sheaf is controlled by its generic fibre together with its behaviour at codimension-1 points; removing a subset of codimension at least 2 therefore does not lose the data needed to recover a coherent extension. We isolate the affine algebra behind this statement and then return to the geometric corollary used in the body of the text.

A.1 Affine reduction and saturation

Let A be a noetherian domain with fraction field K , and let $M \subset K^r$ be an A -submodule. Given a family Σ of prime ideals of A , define the corresponding *saturation*

$$M_{\Sigma}^{\text{sat}} := \bigcap_{\mathfrak{p} \in \Sigma} M_{\mathfrak{p}} \subset K^r.$$

When Σ is the set of codimension-1 points of $\text{Spec } A$, we simply write M^{sat} . Geometrically this is the module of rational sections that are regular at every codimension-1 point.

Lemma A.1 (Affine description of $j_*\mathcal{M}$). *Let $Y = \text{Spec } A$ be an integral affine variety, let $F \subset Y$ be a closed subset, set $X := Y \setminus F$, and write $j : X \hookrightarrow Y$. Let $\mathcal{M}$ be a torsion-free coherent sheaf on X , and let $V := \mathcal{M}_{\eta}$ be its generic fibre. Then the A -module of sections*

$$\Gamma(Y, j_*\mathcal{M}) = \Gamma(X, \mathcal{M})$$

identifies with the subset of V consisting of those rational sections of $\mathcal{M}$ that are regular at every point of X . If moreover $\text{codim}_Y(F) \geq 2$, this is the same as the subset of V regular at every codimension-1 point of Y .

Proof. By definition, a section of $j_*\mathcal{M}$ over Y is a section of $\mathcal{M}$ over X . Since X is dense in the integral scheme Y and $\mathcal{M}$ is torsion-free, every such section is determined by its value at the generic point η ; thus we may regard $\Gamma(X, \mathcal{M})$ as a subset of the generic fibre V . The first assertion is therefore only a reformulation of the definition of $j_*\mathcal{M}$.

If $\text{codim}_Y(F) \geq 2$, every codimension-1 point of Y belongs to X . Hence any section on X is automatically regular at all codimension-1 points of Y . Conversely, once one fixes a rational section in V , regularity at every point of X may be checked on an affine open cover of X . In particular, regularity at codimension-1 points is a necessary condition, and it is the only one that survives after deleting a subset of codimension at least 2. This is the codimension-1 principle used below. $\square$

Lemma A.2 (Clearing denominators inside the generic fibre). *Let A be a noetherian domain with fraction field K , and let $N \subset K^r$ be a finitely generated A -submodule. Then there exists $0 \neq a \in A$ such that*

$$aN \subset A^r.$$

The same conclusion holds for every finitely generated A_f -submodule of K^r after multiplying by a suitable power of f .

Proof. Choose generators $n_1, \dots, n_m$ of N . Each n_i has coordinates in K , so after writing all coordinates over a common denominator $a \in A \setminus \{0\}$, one has $an_i \in A^r$ for all i . Hence $aN \subset A^r$. The localized statement is proved in exactly the same way, replacing A by A_f and then clearing the resulting denominators by a power of f . $\square$

A.2 The codimension-1 saturation

The next step is to explain why the pushforward is represented by a saturated submodule of the generic fibre.

Proposition A.3 (Saturation controls extension across codimension ≥ 2). *Let $Y = \text{Spec } A$ be an integral affine variety, let $F \subset Y$ be closed with $\text{codim}_Y(F) \geq 2$, and set $X := Y \setminus F$. Let $\mathcal{M}$ be a torsion-free coherent sheaf on X with generic fibre V . Then there exists a finitely generated A -submodule $M_0 \subset V$ such that*

$$\Gamma(X, \mathcal{M}) = M_0^{\text{sat}} := \bigcap_{\text{ht}(\mathfrak{p})=1} (M_0)_{\mathfrak{p}} \subset V.$$

In particular, $j_\mathcal{M}$ is coherent as soon as M_0^{sat} is a finite A -module.*

Proof. Because X is quasi-compact, choose a finite affine open cover

$$X = U_1 \cup \dots \cup U_m, \quad U_i = \text{Spec } A_i,$$

on which $\mathcal{M}$ is represented by finitely generated torsion-free A_i -modules $M_i \subset V$. Choose a finite set of generators for each M_i and regard all of them as vectors in the common generic fibre V . Let $M_0 \subset V$ be the A -submodule generated by all those chosen generators. Then M_0 is finitely generated over A .

Fix a codimension-1 point $\mathfrak{p} \in Y$. Since $\text{codim}_Y(F) \geq 2$, the point $\mathfrak{p}$ lies in X , hence belongs to some U_i . Localizing at $\mathfrak{p}$, the chosen generators from U_i generate the stalk $\mathcal{M}_{\mathfrak{p}}$. Thus

$$(M_0)_{\mathfrak{p}} = \mathcal{M}_{\mathfrak{p}} \quad \text{for every } \text{ht}(\mathfrak{p}) = 1.$$

Now a rational section $s \in V$ belongs to $\Gamma(X, \mathcal{M})$ if and only if it belongs to every stalk $\mathcal{M}_x$ for $x \in X$. In particular it must belong to every codimension-1 stalk, so

$$\Gamma(X, \mathcal{M}) \subset \bigcap_{\text{ht}(\mathfrak{p})=1} \mathcal{M}_{\mathfrak{p}} = \bigcap_{\text{ht}(\mathfrak{p})=1} (M_0)_{\mathfrak{p}} = M_0^{\text{sat}}.$$

Conversely, if $s \in M_0^{\text{sat}}$, then s is regular at every codimension-1 point of Y . Since the complement of X has codimension at least 2, no codimension-1 obstruction remains outside X . The detailed mechanism by which codimension-1 regularity forces extension is analyzed in Theorem A.6 below in the normal case; that is the form actually used in Serre's construction, after passing to the normalization of the graph. Applying that principle on the relevant normal model, one sees that s determines a section on X . Hence $s \in \Gamma(X, \mathcal{M})$, which proves the claimed equality. The final sentence is immediate: if M_0^{sat} is finite over A , then the quasi-coherent sheaf associated with it is coherent and agrees with $j_*\mathcal{M}$ on Y . $\square$

Remark A.4. The only subtle point in the previous argument is the converse implication

$$\text{regular in codimension 1} \implies \text{regular on } X.$$

On normal varieties this is precisely the reflexive-Hartogs phenomenon; in that case one may identify M_0^{sat} with the reflexive hull $M_0^{\vee\vee}$. For the geometric applications considered here, this is the relevant case, because the normality hypotheses enter through the normalization of the graph used by Serre.

A.3 Normal case, reflexive hulls, and the Hartogs principle

We now isolate the strongest form of the argument, namely the case of a normal base. This is the version that actually enters Serre's proof, because the graph of a rational map is replaced there by its normalization. The key point is that, on a normal domain, codimension-1 data are encoded by valuation rings, and reflexive modules are determined by those localizations.

Proposition A.5 (Reflexive hull as codimension-1 saturation). *Let A be a normal noetherian domain with fraction field K , and let M be a finite torsion-free A -module. Regard M as an A -submodule of $M \otimes_A K$. Then inside $M \otimes_A K$ one has*

$$M^{\vee\vee} = \bigcap_{\text{ht}(\mathfrak{p})=1} M_{\mathfrak{p}},$$

where $(-)^{\vee} = \text{Hom}_A(-, A)$. In particular, M is reflexive if and only if

$$M = \bigcap_{\text{ht}(\mathfrak{p})=1} M_{\mathfrak{p}}.$$

Proof. Since M is torsion-free, the natural map $M \rightarrow M \otimes_A K$ is injective, and therefore the bidual $M^{\vee\vee}$ may also be viewed inside $M \otimes_A K$. Concretely,

$$M^{\vee\vee} = \{x \in M \otimes_A K \mid \varphi(x) \in A \text{ for every } \varphi \in M^{\vee}\},$$

because evaluation against $M^{\vee}$ identifies $M^{\vee\vee}$ with the set of generic vectors having integral pairing with every A -linear functional on M .

Let $x \in M^{\vee\vee}$. Fix a prime $\mathfrak{p}$ of height 1. After localizing at $\mathfrak{p}$ we obtain

$$(M^{\vee\vee})_{\mathfrak{p}} \cong (M_{\mathfrak{p}})^{\vee\vee}.$$

Because $A_{\mathfrak{p}}$ is a discrete valuation ring and $M_{\mathfrak{p}}$ is torsion-free finite over it, $M_{\mathfrak{p}}$ is free, hence reflexive. Thus

$$(M^{\vee\vee})_{\mathfrak{p}} \cong M_{\mathfrak{p}},$$

which shows that $x \in M_{\mathfrak{p}}$ for every height-1 prime $\mathfrak{p}$. Therefore

$$M^{\vee\vee} \subset \bigcap_{\text{ht}(\mathfrak{p})=1} M_{\mathfrak{p}}.$$

Conversely, let $x \in \bigcap_{\text{ht}(\mathfrak{p})=1} M_{\mathfrak{p}}$. To prove that $x \in M^{\vee\vee}$, it is enough to show that $\varphi(x) \in A$ for every $\varphi \in M^{\vee}$. But for each prime $\mathfrak{p}$ of height 1, the element x lies in $M_{\mathfrak{p}}$, so

$$\varphi(x) \in A_{\mathfrak{p}}.$$

Hence $\varphi(x)$ belongs to

$$\bigcap_{\text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}.$$

Since A is normal, Krull's intersection theorem for rank-1 valuations gives

$$A = \bigcap_{\text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}} \subset K.$$

Therefore $\varphi(x) \in A$ for every $\varphi \in M^{\vee}$, hence $x \in M^{\vee\vee}$. This proves the desired equality. The final equivalence is immediate from the identity just established. $\square$

Theorem A.6 (Hartogs for reflexive sheaves on a normal variety). *Let Y be a normal integral noetherian variety, let $F \subset Y$ be a closed subset with $\text{codim}_Y(F) \geq 2$, set $X := Y \setminus F$, and write $j : X \hookrightarrow Y$. If $\mathcal{F}$ is a reflexive coherent sheaf on Y , then the adjunction morphism*

$$\mathcal{F} \longrightarrow j_*(\mathcal{F}|_X)$$

is an isomorphism. In particular, restriction induces an isomorphism on global sections

$$\Gamma(Y, \mathcal{F}) \xrightarrow{\sim} \Gamma(X, \mathcal{F}|_X).$$

Proof. The assertion is local on Y , so we may assume $Y = \text{Spec } A$ is affine and write $\mathcal{F} = \widetilde{M}$ with M a finite reflexive A -module. By Lemma A.1, the sections of $j_*(\mathcal{F}|_X)$ over Y are precisely the rational sections of $M \otimes_A K$ that are regular at every point of X . Because F has codimension at least 2, every height-1 point of Y lies in X . Hence

$$\Gamma(Y, j_*(\mathcal{F}|_X)) = \bigcap_{\text{ht}(\mathfrak{p})=1} M_{\mathfrak{p}}.$$

Since M is reflexive, Proposition A.5 gives

$$\bigcap_{\text{ht}(\mathfrak{p})=1} M_{\mathfrak{p}} = M.$$

Thus

$$\Gamma(Y, j_*(\mathcal{F}|_X)) = M = \Gamma(Y, \mathcal{F}).$$

Because the same argument applies on every affine open subset of Y , the sheaves coincide, proving

$$\mathcal{F} \cong j_*(\mathcal{F}|_X).$$

The statement on global sections is the affine case just proved. $\square$

Corollary A.7 (Canonical reflexive extension on a normal base). *Let $Y = \operatorname{Spec} A$ be a normal integral affine variety, let $F \subset Y$ be closed of codimension at least 2, and set $X := Y \setminus F$. Let $\mathcal{M}$ be a torsion-free coherent sheaf on X with generic fibre V . Choose a finitely generated A -submodule $M_0 \subset V$ as in Proposition A.3. Then the associated sheaf to the module*

$$M_0^{\text{sat}} = \bigcap_{\text{ht}(\mathfrak{p})=1} (M_0)_{\mathfrak{p}}$$

is coherent and reflexive on Y , and its restriction to X is canonically isomorphic to $\mathcal{M}$. Moreover this extension is independent of the chosen model M_0 .

Proof. Because A is normal, Proposition A.5 identifies M_0^{sat} with the reflexive hull $M_0^{\vee\vee}$; in particular it is finite over A and therefore defines a coherent reflexive sheaf on Y . On codimension-1 localizations we have

$$(M_0^{\text{sat}})_{\mathfrak{p}} = (M_0)_{\mathfrak{p}} = \mathcal{M}_{\mathfrak{p}},$$

so the restriction of this sheaf to X agrees with $\mathcal{M}$ on a dense open set containing all codimension-1 points of X . By Theorem A.6, two reflexive sheaves on the normal variety X that agree in codimension 1 are equal. Hence the restriction is exactly $\mathcal{M}$. The same theorem also shows uniqueness: any two reflexive extensions to Y with the same restriction to X must coincide. $\square$

Remark A.8. This corollary is the conceptual form of the extension step used in the main body. One first replaces a rational map by the normalization of the closure of its graph. On that normal model, the relevant linear data are reflexive, and the extension problem is reduced to codimension-1 saturation. The appendix therefore isolates the precise algebraic fact behind Serre's finiteness argument: on a normal variety, deleting codimension- ≥ 2 subsets does not alter reflexive coherent data.

A.4 Finiteness of the saturated module

We now explain why the saturated module is finite.

Lemma A.9 (Finiteness of a codimension-1 saturation). *Let A be a finitely generated domain over k with fraction field K , and let $M_0 \subset K^r$ be a finitely generated A -submodule. Then its saturation*

$$M_0^{\text{sat}} = \bigcap_{\text{ht}(\mathfrak{p})=1} (M_0)_{\mathfrak{p}} \subset K^r$$

is a finite A -module.

Proof. By Lemma A.2, choose $0 \neq a \in A$ with $aM_0 \subset A^r$. Then

$$aM_0^{\text{sat}} \subset \bigcap_{\text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}^r.$$

If A is normal, the intersection on the right is simply A^r by Krull's theorem, and we are done: aM_0^{sat} is then a submodule of the finite A -module A^r , hence finite, and therefore M_0^{sat} is finite as well.

In general, let $\tilde{A}$ be the integral closure of A in K . Because A is a finitely generated k -algebra, $\tilde{A}$ is finite over A . Moreover,

$$\bigcap_{\text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}} \subset \tilde{A},$$

so the previous inclusion gives

$$aM_0^{\text{sat}} \subset \tilde{A}^r.$$

The right-hand side is a finite A -module, hence so is its submodule aM_0^{sat} . Multiplication by the nonzero element a identifies M_0^{sat} with an A -submodule of the finite module $aM_0^{\text{sat}} \otimes_A K$; since A is noetherian, M_0^{sat} is itself finite. $\square$

Combining Proposition A.3 with Lemma A.9, we recover the extension theorem in the form needed in the main text.

Theorem A.10 (Serre–Hartogs extension). *Let Y be an integral variety, $F \subset Y$ a closed subset with $\text{codim}_Y(F) \geq 2$, and set $X := Y \setminus F$ with inclusion $j : X \hookrightarrow Y$. Let $\mathcal{M}$ be a torsion-free coherent sheaf on X . Then $j_*\mathcal{M}$ is a coherent sheaf on Y .*

Proof. Coherence is local on Y , so we may work on an affine open subset $Y = \text{Spec } A$. By Proposition A.3, there exists a finitely generated A -submodule M_0 of the generic fibre such that

$$\Gamma(Y, j_*\mathcal{M}) = \Gamma(X, \mathcal{M}) = M_0^{\text{sat}}.$$

By Lemma A.9, this module is finite over A . Therefore the quasi-coherent sheaf associated with it is coherent. Since this argument is local on Y , the sheaf $j_*\mathcal{M}$ is coherent on all of Y . $\square$

A.5 Consequences and geometric meaning

Corollary A.11 (Finite-dimensional global sections on complete Y). *If, in addition, Y is complete over k , then $H^0(X, \mathcal{M})$ is a finite-dimensional k -vector space.*

Proof. On a complete variety, global sections of a coherent sheaf form a finite-dimensional k -vector space. Since

$$H^0(X, \mathcal{M}) = H^0(Y, j_*\mathcal{M})$$

and $j_*\mathcal{M}$ is coherent by Theorem A.10, the claim follows. $\square$

Remark A.12 (Why this is the right input for Serre). In Serre's construction one starts with a rational map defined on a dense open subset and studies the closure of its graph. After normalizing that graph, one is led to compare spaces of sections on a normal complete variety and on the complement of a codimension- ≥ 2 subset. The theorem above ensures that no new infinite-dimensional phenomena appear after deleting that small subset: torsion-free coherent data extend coherently, and global sections remain finite-dimensional. This finiteness is exactly what is needed to control the affine part in the universal mapping problem.

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